

Global Considerations in Kidney Disease: The Far East

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CHAPTER OUTLINE

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KEY POINTS

- Along with rapid economic development, diabetic nephropathy is fast becoming the leading cause of chronic kidney disease and end-stage kidney disease in many Asian countries. Data from major kidney biopsy registries show that immunoglobulin A nephropathy (IgAN) is the leading diagnosis among primary glomerular diseases, and lupus nephritis is the leading diagnosis among secondary glomerulonephritides.
- Advances in the management of lupus nephritis over the past few decades have resulted in improved clinical outcomes including reduced disease flares and increased kidney and patient survival rates, and the advent of novel treatments has facilitated individualized therapeutic intervention.
- Results from major clinical trials in IgAN highlight the potential risks of high-dose oral corticosteroid treatment and the importance of judicious balance between potential benefit versus risk in treatment decisions. Data demonstrate the efficacy of enteric-release budesonide.
- Aristolochic acid nephropathy illustrates the importance of environmental toxins in causing acute kidney injury or chronic kidney disease, especially when the etiology is obscure. Recent data implicate air pollutants in the pathogenesis of idiopathic membranous nephropathy and chronic kidney disease.
- In contrast to Caucasian populations in which granulomatosis with polyangiitis and antiproteinase 3 antibodies are more common, antineutrophil cytoplasmic antibody-associated vasculitis in East and Northeast Asia is characterized by a high percentage of patients presenting with microscopic polyangiitis and antimyeloperoxidase antibodies and more chronic lesions in the kidney biopsy.
- In the current era with effective nucleoside/tide therapy for hepatitis B, the short- and medium-term survival of hepatitis B surface antigen (HBsAg)-positive kidney transplant recipients is comparable with those without hepatitis B virus infection.

The term “Far East” refers to East Asia (including Northeast Asia), part of North Asia, and Southeast Asia (Fig. 78.1). In addition to its vast geographic extent and big population, Asia is characterized by marked variations in socioeconomic development, ethnicity, and culture, both between countries and within big countries. Such heterogeneity is associated with marked differences in epidemiology of kidney diseases including both causes of acute kidney injury (AKI) and

chronic kidney disease (CKD), and access to health care, the latter with direct implications on clinical outcomes, especially patient and kidney survival rates. The disease pattern in developed areas such as Japan, Korea, Singapore, and Hong Kong largely resembles that of the West, with diabetic nephropathy being the major cause of end-stage kidney disease (ESKD) in adults, immunoglobulin A (IgA) nephropathy being the leading primary renal parenchymal



Fig. 78.1 Countries and territories that constitute the Far East. (Modified from Wikipedia, https://en.wikipedia.org/wiki/Geography_of_Asia.) Map lines delineate study areas and do not necessarily depict accepted national boundaries.

disease, and lupus nephritis being an important cause of acute glomerulonephritis and CKD in both adults and children. Some conditions are more common in Asia compared with other parts of the world (e.g., IgA nephropathy [IgAN], lupus nephritis, chronic hepatitis B, and herbal medicine-associated AKI or CKD). Conditions that are typically found in the tropics or subtropics, such as malaria or dengue fever, can present as imported cases in other Asian countries. AKI associated with infections such as leptospirosis, scrub typhus, or hemorrhagic fever with renal

syndrome (HFRS) due to hantavirus is present in some countries. The massive volume of international travel and commuting by humans and traffic of commodities between nations, coupled with climate change and global warming, have contributed to the spread of certain diseases such as dengue fever to previously less afflicted areas.

While kidney diseases, especially AKI and CKD, constitute a major public health problem with increasing cost to communities globally, the data on disease varieties and burden, as well as on the kidney health care structure and

delivery systems for treatment and prevention, have been fragmented. Also, despite the huge disease burden and societal cost, overarching health care policies targeting the control and intervention of kidney diseases are lacking in many countries. The International Society of Nephrology Global Kidney Health Atlas project, a multinational survey launched in 2016 and with its third edition published in 2023, has addressed the knowledge gap and provides valuable insights, on a global scale, on the impact of kidney diseases on individuals and societies and the state of affairs with regard to health care policies, human and facility resources, and access to treatments.¹ As expected, the findings showed marked diversity in health care funding, nephrologist density, and institutional kidney care provision within Asia and still considerable knowledge gaps in some Asian countries. For example, data on patients receiving maintenance dialysis are available only in countries or areas with national health insurance or renal registries such as Japan, Korea, Taiwan, Singapore, and Hong Kong, and few countries have data on AKI or nondialysis CKD. Renal replacement therapies are costly, and hemodialysis (HD) facilities may not be immediately available to patients with AKI in rural areas. The outcome of patients with AKI would thus be adversely affected due to financial disadvantage or inconvenient social settings.

Health expenditure and health care funding models, which have a direct impact on the access to health care, vary widely between Asian countries. Economic development varies markedly within the Far East region, ranging from high-income economies, such as Japan, South Korea (Republic of Korea), Singapore, Hong Kong, and Taiwan, to low- or lower-middle income economies such as North Korea (Democratic People's Republic of Korea), the Philippines, Indonesia, Vietnam, Cambodia, Lao People's Democratic Republic, Myanmar, and Mongolia.² Among Asian countries, Japan has the highest health expenditure, amounting to 10.9% of gross domestic product (GDP) or 43 trillion yen in 2020, compared with 18.3% or US\$188 billion in the United States in 2021. Health care systems in places such as Japan, South Korea, Singapore, and Taiwan are based on national insurance and are well developed, while the health insurance coverage is being expanded in other areas such as China.

TRADITIONAL MEDICINE

Many countries in Asia have a long history of traditional systems of medicine, and the use of traditional herbal medicine is common among Chinese in different countries and in other Asian countries such as Korea, Thailand, and Japan. Traditional medicines have served a significant role in providing health care access to patients in less developed regions and in relatively low-income economies. For example, it has been reported that traditional Chinese medicine accounted for >20% of health care services and treated approximately 200 million people annually in China.³ Traditional medicine is covered to variable extents by national health insurance schemes in China, Taiwan, South Korea, and Japan, and there are formal education systems for traditional medicine in China and Korea.⁴ Regulatory systems are in place in some areas, such as the State Traditional

Chinese Medicine Administration in China, the Ministry of Health and Welfare in Korea, the Ministry of Health and Welfare in Taiwan, and the Department of Health in Hong Kong, which include compliance requirements with regard to good manufacturing practice.⁵ In a 48-week clinical trial that included 148 patients with type 2 diabetes and eGFR 30 to 90 mL/min/1.73 m², add-on treatment with Rehmannia-6-based Chinese medicine was associated with reduced rate of eGFR decline (−2.0, compared with −4.7 mL/min/1.73 m²/year, $P = 0.04$) without alteration of proteinuria level.⁶ Due to the complexities in plant taxonomy and the components that are related to medicinal effects or toxicities, adverse effects related to intentional or inadvertent exposure to herbal medicines occur. Nephrotoxicity of *Tripterygium wilfordii*, a traditional Chinese medicine used in the treatment of lupus, rheumatoid diseases, and psoriasis, is well recognized,⁷ and the etiology, modes of exposure, and pathogenic mechanisms in Aristolochic acid (AA) nephropathy serve as illustrative examples.⁸ A study in Taiwan reported a 20% increase in the risk of developing CKD in subjects who took herbal medicine regularly.⁹ CKD associated with traditional medicine use was associated with increased blood cadmium and 5-methyl-2-deoxycytidine level, a marker of global DNA methylation.¹⁰ There was also an association between blood levels of lead and cadmium, highlighting the possible impact of environmental heavy metal exposure from traditional medicines on kidney health. It is foreseeable that traditional medicine will continue to have a significant role in the healthcare systems of many countries in Asia. More research to understand the specific mechanisms of action of individual items or ingredients, and regulatory measures to ensure safety, are both important.

This chapter discusses the kidney diseases that are more common, or of special interest, in the Far East, focusing on recent data and developments.

ACUTE KIDNEY INJURY

AKI can be induced by a diverse range of endogenous or environmental factors. AKI is estimated to affect >13 million people per year, although few countries maintain a record or have a registry for AKI patients.¹¹ AKI is an important cause of hospital admission in Asian countries, and the leading causes of AKI vary according to the socioeconomic development status and the living standard of the general population in different countries.^{11,12} In areas with less developed building infrastructure and where the majority of the population is engaged in agriculture, community-acquired AKI is common and the causes include infections and animal or plant toxins (Table 78.1).^{13,14} Some of the causes are related to culture or folklore (e.g., accidental exposure to toxin in the grass carp fish bladder used by merchants to replace the more costly snake gallbladder or ingestion of raw or dried cyprinid fish gallbladder as a health tonic or for the treatment of rheumatism).¹⁵ The development of AKI is sometimes accompanied by hepatotoxicity and myopathy due to the 5- α -cyprinol sulfate nephrotoxin/hepatotoxin.¹⁶ Another example of great public health importance is the exposure to AA present in herbs used in traditional medicine, which leads to not only CKD but also AKI and multifocal urological malignancies.^{17,18} In more developed high-income countries such as Japan,

Table 78.1 Secondary Renal Diseases Presenting as Acute Kidney Injury in East, Southeast, and Northeast Asia

Causes	Examples
Autoimmunity	• Lupus nephritis • ANCA vasculitis • Antiglomerular basement membrane disease • Miscellaneous glomerulonephritis
Infections	• Leptospirosis • Dengue fever • Hantaviruses (hemorrhagic fever with renal syndrome) • Scrub typhus • Malaria • Severe fever with thrombocytopenia syndrome virus
Toxins	• From plant (e.g., aristolochic acid and djenkolic acid) • From animal or insect (e.g., snake venom and grass carp fish gallbladder) • Chemical (e.g., ketamine abuse presenting with CKD or AKI due to obstructive uropathy)

AKI, Acute kidney injury; ANCA, antineutrophil cytoplasmic antibody; CKD, chronic kidney disease.

Korea, or Singapore, causes such as toxin- or infection-associated AKI are much less common, while AKI is often a complicating factor in hospitalized critically ill patients with cardiovascular compromise, surgical conditions, and severe infections. In large countries with rapid economic development, such as China, the epidemiology of AKI varies according to geographic and topographic location.¹⁹ Knowledge of locally prevalent causes of AKI is thus crucial to timely diagnosis and therapeutic intervention.

CHRONIC KIDNEY DISEASE

CKD is a growing global health burden with a prevalence rate of >9%, and Asia is no exception.²⁰⁻²³ A report on the prevalence and temporal trends of CKD, based on the Global Burden of Diseases Study 2019, showed that in 2019 there were 3.16 million deaths and 76.5 million disability-adjusted life years attributable to kidney disease, an increase of 101.1% and 81.7%, respectively, compared with 1990 with apparently greater increases in countries of lower sociodemographic index.²⁴ A report in 2015 showed that, in the year 2010, 218 patients per million population (pmp) were receiving dialysis and 232 patients pmp were receiving renal replacement therapy (RRT) in Asia. In comparison, 1839 patients pmp were on renal replacement therapies in the United States in the same period, increasing to 6437 patients pmp among Black individuals and 2361 patients pmp among Asian individuals in 2019.^{25,26} A meta-analysis reported in 2016 showed that the prevalence rate of stage 1 to 5 CKD was approximately 13% in China, Japan, Korea, and Taiwan, whereas that for stage 3 to 5 CKD was approximately 10% to 12%.²¹

A systematic analysis on global disease burden reported 1.2 million deaths from CKD in 2017 and that the age-standardized CKD prevalence rate per 100,000 population in 2017 was 11,079 in Malaysia, 11,049 in the Philippines, 10,323 in Brunei, 10,292 in Thailand, 10,107 in Vietnam, 9333 in Singapore, 8404 in Japan, 8145 in Taiwan, 7180 in China, and 7103 in S Korea, compared with 9529 in India, 8144 in the United States, and 6982 in Australia.²⁷ An earlier study in China reported a CKD prevalence rate of 10.8%, implying that approximately 120 million adults in China had reduced kidney function and/or albuminuria.²⁸ A report from South Korea in 2016, based on the results of the Korean National Health and Nutritional Examination Survey in 2011–2013 that involved 23,280 subjects, showed that the prevalence rate of CKD as defined by urine albumin-to-creatinine ratio ≥ 30 mg/g or estimated glomerular filtration rate (eGFR) < 60 mL/min/1.73 m² with the Chronic Kidney Disease-Epidemiology Collaboration (CKD-EPI) equation was 8.2% in adults at or above the age of 20 years, with respective proportions in stages 1, 2, 3a, 3b, and 4 to 5 being 3.0%, 2.7%, 1.9%, 0.4%, and 0.2%.²⁹ Factors associated with increased risk of CKD included older age, diabetes, hypertension, cardiovascular disease, body mass indexes ≥ 25 kg/m² or below 18.5 kg/m², and residence in a rural area. Urbanization and economic development are associated with increasing prevalence rates of diabetes mellitus and hypertension, important contributors to the increasing burden of CKD in high-income countries or metropolitan areas. A cross-sectional study based on data from a private health-check agency on 38,093 subjects (85.6% younger than the age of 60 years) from 10 most economically developed major cities in People's Republic of China detected 3837 subjects with CKD (10.1%), and 204 of these subjects had eGFR below 60 mL/min/1.73 m².³⁰

The KoreaN Cohort Study for Outcomes in Patients With Chronic Kidney Disease (KNOW-CKD) investigated the factors that impact the progression of CKD, based on data from 2238 subjects enrolled in 2011 to 2015.^{31,32} The age of the cohort was 53.7 ± 12.2 years and 61.2% were men. eGFR was 50.5 ± 30.3 mL/min/1.73 m², and lower eGFR values were associated with older age, comorbidities, higher systolic blood pressure, greater pulse pressure, lower income level, and lower education attainment. Glomerulonephritis, diabetic nephropathy, hypertensive nephropathy, and polycystic kidney disease accounted for 36.2%, 23.2%, 18.3%, and 16.3% of cases, respectively. The overall prevalence of previous cardiovascular disease was 14.4%, and the rate was highest in those with diabetes.³² While early diagnosis is important for timely intervention to delay disease progression, patients are often not diagnosed since they are asymptomatic until CKD reaches the late stages of 4 or 5. One study (REVEAL-CKD) based on eGFR values from electronic medical records and/or insurance claims databases showed that a high proportion of patients with CKD stage 3 did not have this diagnosis entered into their records—92.1% in Japan (with CKD stage 3 diagnosed in 3.1% of the surveyed population of 2.9 million subjects), 95.5% in France (CKD stage 3 in 3.3% of 0.6 million subjects), and 61.6% in the U.S. Explorys Linked Claims and Electronic Medical Records Data database (CKD stage 3 in 7.6% of 3.3 million subjects), suggesting that

their CKD might be undiagnosed or not accorded enough significance.³³ In this context, a multidisciplinary approach that involves nurses, dieticians, and other health care personnel may provide improved care to patients with CKD. A retrospective study that included 3015 Japanese patients with CKD stages 3 to 5 showed that multidisciplinary care was associated with a slower rate of eGFR decline, regardless of the cause of CKD.³⁴

The relationship between kidney function and survival was investigated in the population-based China National Survey of Chronic Kidney Disease study, which recruited adult subjects in 13 provinces starting from 2007.²⁸ Data from 47,204 subjects over a median follow-up duration of 6.1 years showed that all-cause and cardiovascular mortality increased with increasing severity of CKD and also albuminuria.³⁵ Baseline albumin-to-creatinine ratio ≥ 300 mg/g was associated with a hazard ratio (HR) of 2.07 for all-cause mortality and 2.32 for cardiovascular mortality compared with subjects with a ratio < 30 mg/g. Interestingly, reduced eGFR was not identified as a death risk predictor in multivariate analysis, although there was interaction between eGFR and age.³⁵

AIR POLLUTION AND CHRONIC KIDNEY DISEASE

On the basis of more than 70,000 biopsy-proven kidney diseases in 282 cities in China from 2004 to 2014, investigators in China showed that the leading diagnoses were IgAN (28.1%) and membranous nephropathy (23.4%) and that the incidence of membranous nephropathy increased by 13% annually over the period while the other glomerulopathies remained stable.³⁶ Importantly, the odds of occurrence of membranous nephropathy were associated with the severity of air pollution, with 14% higher odds for each $10 \mu\text{g}/\text{m}^3$ increase in PM_{2.5} (fine particles with an aerodynamic diameter of ≤ 2.5 mm) concentration in regions with PM_{2.5} $> 70 \mu\text{g}/\text{m}^3$. A high percentage of patients with idiopathic membranous nephropathy has anti-PLA2R antibodies in their serum. While PLA2R is expressed in podocytes, it is also present in alveolar macrophages, airway epithelial cells, and submucosal epithelial cells. It has been postulated that exposure to air pollutants could induce lung inflammation or injury, which leads to the introduction of PLA2R into systemic circulation and initiation of an autoimmune response.³⁷ A study in Taiwan that included $> 100,000$ adult subjects over the period 2001 to 2014 showed an incidence rate for CKD of 6.24 per 1000 person-years, which was higher in participants with greater long-term exposure to particulate matter in the air (HR 1.11 and 1.15 for the top fourth and fifth quintiles, respectively, compared with the lowest first quintile) with a concentration-response trend.³⁸ Similarly, a retrospective nationwide analysis of health check-up records from the period 2012 to 2017 that included approximately 3 million adults showed a high average PM_{2.5} concentration of $78.7 \pm 22.5 \mu\text{g}/\text{m}^3$ and increased odds ratio of CKD prevalence related to PM_{2.5} exposure in the previous year.³⁹ The relationship was applicable in all five PM_{2.5} components tested (ammonium, nitrate, organic matter, black carbon, and sulfate), with the latter showing the highest contribution. Various pathogenic mechanisms have been postulated for

the association between air pollution and CKD including renal inflammation, oxidative stress, apoptosis, DNA damage, and autophagy of kidney cells.⁴⁰

CARDIOVASCULAR COMPLICATIONS IN CKD

Morbidity and mortality in patients with CKD demonstrate considerable regional variations. Patients on dialysis have increased risk for cardiovascular morbidity and mortality. A systematic analysis up to 2017 reported that globally 1.4 million deaths from cardiovascular disease were attributable to CKD. More recent results showed that almost 7% of total cardiovascular disease burden could be attributable to CKD, and CKD has surpassed ischemic heart disease as the most important factor for excessive mortality.^{24,27} Interestingly, the prevalence rate of cardiovascular complications varies by geographic region. Results from 16,560 HD patients from the Dialysis Outcomes and Practice Patterns Study (DOPPS) phase 4 (2009–2011) and phase 5 (2012–2015) cohorts showed that the rate of major adverse cardiovascular events (MACE) was highest in North America (19.4 per 100 patient-years), followed by Europe (17.4), and it was lowest in Japan (7.5).⁴¹ In this regard, Japan and the United States are both high-income countries with similar CKD prevalence (age-standardized prevalence rate of approximately 8% in 2017), while age-standardized death rate attributable to CKD in the United States was double that in Japan (14.6 and 7.5 per 100,000 respectively).²⁷ For predialysis CKD patients, analysis of data from the Chronic Renal Insufficiency Cohort (CRIC) and CKD Japan Cohort (CKD-JAC) showed that Japanese patients had significantly higher left ventricular ejection fraction (65% vs. 54%) and lower left ventricular mass index (46.6 vs. 55.7 g/m²) compared with patients from the United States, which in turn were significantly associated with cardiovascular events during follow-up, and patients in the CRIC cohort showed higher risks of cardiovascular disease and death (adjusted HR 3.66 and 4.69, respectively).⁴²

HEALTH CARE ECONOMICS AND FINANCING MODELS FOR CHRONIC KIDNEY DISEASE

Health care economics has a major impact on access to clinical care and the choice of therapies. This is particularly relevant to CKD care and renal replacement therapies, which are costly, especially in Asia where the rate of organ transplantation is low and many patients remain on dialysis for decades.^{43,44} Not surprisingly, large treatment gaps with regard to RRT have been noted in low-income countries in Asia.²⁵ Data from the International Society of Nephrology Global Kidney Health Care Status survey showed a variety of financing models ranging from total government funding to a mixture of public and private facilities for dialysis service in North and East Asia, with marked variations in the availability of services for CKD monitoring and management at the primary care level.^{43,45} In addition, few countries have a registry for nondialysis CKD (or AKI) patients.^{43,46} Different approaches have been adopted to address distinct priorities, circumstances, and challenges in kidney disease care in different countries.⁴⁷ More mature care models, such as those

in Japan, Taiwan, and South Korea, encompass interventions and therapies at different phases of CKD progression from early intervention to kidney replacement therapies.

The challenge to health care financing for CKD is compounded when economic development often brings about increasing life expectancy and a growing prevalence of CKD.²⁵ Heterogeneities in health care spending, the developmental status of nephrology care and personnel, and sociocultural background result in marked variations in practice across Asian countries. For example, the health expenditure as a percentage of GDP in 2014 was 10.2% in Japan, highest among Asian countries, and was 7.4% in Korea, 5.5% in China, 4.7% in the Philippines, and 4.1% in Thailand, compared with 9.4% in Australia, 9.1% in the United Kingdom, 11.3% in Germany, and 17.1% in the United States.⁴⁸ National health care insurance in places such as Japan, Taiwan, Korea, and Singapore provides coverage for dialysis therapy of residents. In places such as Hong Kong, Thailand, and Vietnam, economic or facility considerations have led to peritoneal dialysis (PD) being promoted or preferred to HD for the treatment of patients with ESKD as public health care policies.⁴⁹⁻⁵⁴ Similarly, government-sponsored reimbursement schemes have led to an increase in PD utilization in some provinces in China, so the number of PD patients in China increased from 37,942 in 2012 to 62,589 in 2015.^{55,56} Twice-weekly or incremental HD has also been used in patients of small body size or with residual renal function to limit health care expenditure.^{57,58}

In Japan, maintenance HD for patients with ESKD started in 1963.⁵⁹ A report from the Japanese Society for Dialysis Therapy Renal Data Registry in 2015, which covered 304,856 prevalent dialysis patients in 2011 (97% of HD), showed a progressive increase in the number of dialysis patients over the years, primarily due to an increased number of patients older than the age of 65 years. The mean age of prevalent dialysis patients

was 66.6 years in 2011, compared with 46.2 years in 1981.^{60,61} Diabetes mellitus was the leading cause of ESKD, accounting for approximately 40% of incident patients and 45% of prevalent cases. By contrast, glomerular diseases accounted for 75% of incident patients in 1983, decreasing to around 35% in 2011. Cardiovascular disease was the leading cause of mortality, accounting for around 40%. In addition, approximately 30% of dialysis patients were older than the age of 75 years and 8% of patients had been on dialysis for >20 years.⁶⁰

The current kidney transplantation rate only meets approximately 10% of the demand globally and is lowest in Asia.^{62,63} Attempts to overcome some of the challenges have resulted in the development of kidney transplantation across the ABO blood group barrier,⁶⁴ exchange donor kidney transplantation,⁶⁵ improvements in the management of hepatitis B after kidney transplantation so that hepatitis B virus (HBV)-infected renal failure patients can be included for kidney transplantation,^{66,67} and the use of hepatitis B surface antigen (HBsAg)-positive kidney donors.⁶⁸

SPECIFIC KIDNEY DISEASES

IgA NEPHROPATHY

IgAN is the most common primary glomerulonephritis worldwide (see also [Chapter 33](#)), and the condition is common in Asia ([Fig. 78.2](#)).⁶⁹ Data from renal biopsy registries in China showed that IgAN accounted for >45% of primary glomerular diseases.^{70,71} The pathogenesis of IgAN is related to overproduction of aberrantly glycosylated IgA1 and the formation of immune complexes with autoantibodies that recognize the undergalactosylated IgA1 molecule.⁶⁹ Downstream pathways lead to activation of mesangial cells and overproduction of cytokines, chemokines, and other

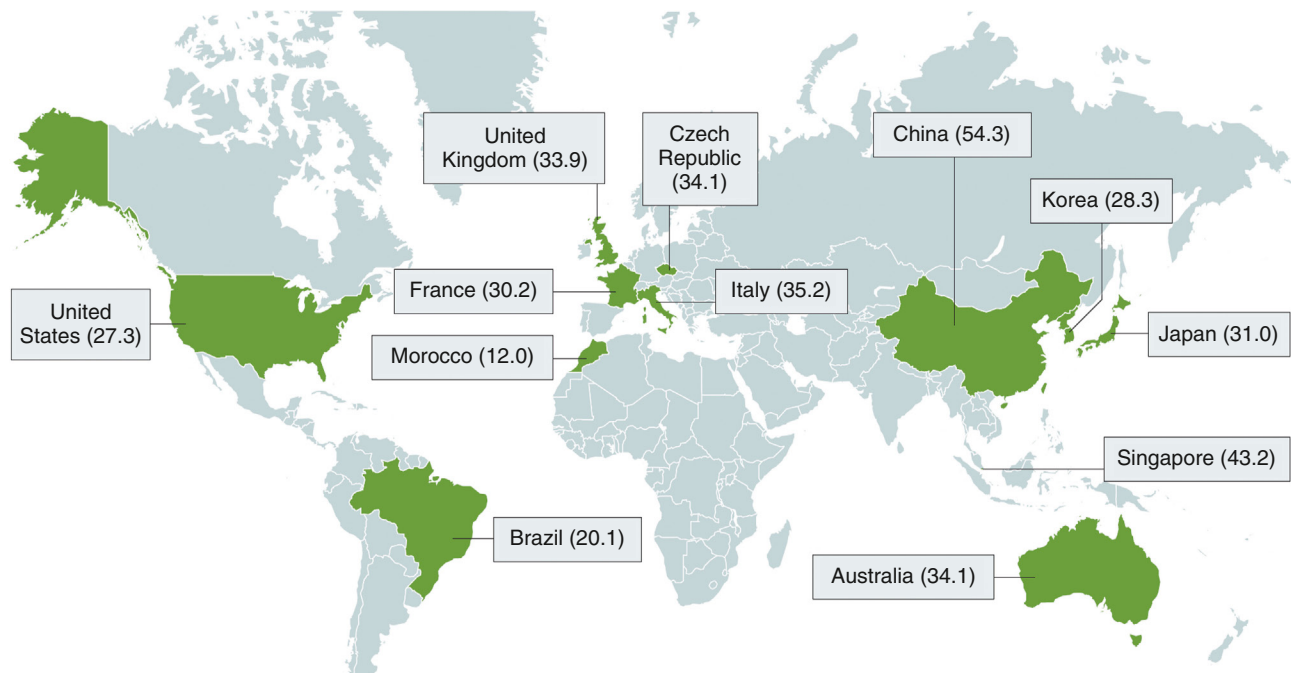


Fig. 78.2 Prevalence of immunoglobulin A nephropathy as indicated by percentage of biopsy-proven primary glomerulonephritis in different parts of the world. (Reprinted by permission from MacMillan Publishers Ltd. From Lai KN, Tang SCW, Schena FP, et al. IgA nephropathy. *Nat Rev Dis Primers*. 2016;2:16001.) Map lines delineate study areas and do not necessarily depict accepted national boundaries.

proinflammatory and profibrotic mediators locally, resulting in inflammation and fibrosis.

Susceptibility loci and variants for IgAN have been identified in both HLA and non-HLA gene regions, the latter often related to immune effector or mediator cells.^{72,73-75} Studies of familial IgAN in Chinese using whole-exome sequencing identified six deleterious variants in four genes, namely defensin alpha 4 (*DEFA4*), MYC target 1 (*MYCT1*), caspase recruitment domain family member 8 (*CARD8*), and zinc-finger protein 543 (*ZNF543*).⁷⁶ Disease susceptibility has also been associated with genes involved in antigen processing and presentation such as the major histocompatibility complex locus or the mucosal defense system (the *DEFA* gene cluster), as well as the alternative complement pathway.^{72,77}

Clinical presentation of IgAN is variable, ranging from asymptomatic microscopic hematuria with or without proteinuria or renal impairment to rapidly progressive renal impairment associated with abundant proliferative features and crescent formation in the glomeruli. Overall, about 30% to 40% of patients reach ESKD in 20 to 30 years after initial presentation, but the prognosis and rate of progression varies considerably between patients.⁶⁹ Analysis of pooled data from 3096 patients, including 700 patients from Japan and 1017 from China, showed that crescents predicted a higher risk of the combined renal endpoint of $\geq 50\%$ eGFR decline or ESKD. Having crescents in at least one-sixth or one-fourth of glomeruli was associated with an HR of 1.63 or 2.29, respectively.⁷⁸ The T score in the Oxford classification and the Japanese Histologic Classification score have been shown to be associated with renal outcome in a series that included 86 Japanese patients.⁷⁹ Investigators in Singapore showed that mesangial hypercellularity and tubular atrophy/interstitial fibrosis scores (M1 and T1/T2 lesion) of the Oxford classification independently predicted ESKD over long-term follow-up.⁸⁰

Treatment decisions take into account clinical and histologic parameters. Optimal treatment of hypertension, renin-angiotensin-aldosterone system (RAAS) inhibition or blockade, sodium-glucose cotransporter-2 inhibitors, endothelin and angiotensin II receptor antagonists, and possibly specific nonsteroidal mineralocorticoid receptor antagonist, are standard renoprotective measures. A study in Japan showed that mizoribine treatment conferred no benefit on proteinuria.⁸¹

Results from the Supportive Versus Immunosuppressive Therapy for the Treatment of Progressive IgA Nephropathy (STOP-IgAN) trial, which included 162 Caucasian patients in Germany, showed that the addition of immunosuppressive therapy using corticosteroids with or without cyclophosphamide or azathioprine in patients who showed persistent proteinuria of at least 0.75 g/day while on standard of care reduced proteinuria but did not reduce the rate of eGFR decline, and immunosuppression was associated with more infections (resulting in one death), impaired glucose tolerance, and weight gain.⁸² The Therapeutic Evaluation of Steroids in IgA Nephropathy Global (TESTING) trial investigated the efficacy and safety of oral methylprednisolone in 503 patients with IgAN with proteinuria not lower than 1 g/D despite standard therapy.⁸³ After 262 patients (251 patients in China) were randomized, an excess of serious infections was identified, leading to methylprednisolone

dose reduction from 0.6 to 0.8 mg/kg daily (maximum 48 mg/day) for 2 months and then tapered by 8 mg per day each month with a total treatment duration of 6 to 8 months originally to 0.4 mg/kg per day (maximum, 32 mg/day) for 2 months, followed by dose tapering by 4 mg per day each month for a total of 6 to 9 months and addition of prophylaxis against pneumocystis. Over 4.2 years of follow-up, 40% eGFR drop or kidney failure occurred in 28.8% in the methylprednisolone group compared with 43.1% in the placebo group (HR 0.53, $P < 0.001$). More recently, targeted-release budesonide aiming to act on the ileocecal Peyer patches was shown to be effective in reducing proteinuria and eGFR preservation in phase 2 and phase 3 trials.^{84,85} In a trial that included 176 Chinese patients with IgAN and cellular/fibrocellular crescents in $\geq 10\%$ of glomeruli or endocapillary hypercellularity or glomerular necrosis, treatment with high-dose prednisone alone was comparable with mycophenolate combined with reduced-dose prednisone with regard to proteinuria reduction, and approximately 50% of patients in each group achieved complete renal remission with the latter group showing fewer treatment-related adverse effects.⁸⁶ The data to date are insufficient to conclude that race or ethnicity has an impact on the response of IgAN to immunosuppressive treatments.

Tonsillar lymphocytes of patients with IgAN showed increased production of dimeric IgA1 with altered glycosylation.^{87,88} Animal experiments demonstrated BAFF-driven pathobiont-specific IgA production in mouse kidney.⁸⁹ However, the role of tonsillectomy in the management of IgAN remains controversial. Tonsillectomy, with or without corticosteroids, has been associated with more favorable renal outcome^{90,91} and is considered a standard of care treatment for IgAN in Japan.⁹²⁻⁹⁵ By contrast, analysis of data from the VALIGA cohort of 1147 European patients with IgAN showed no association between tonsillectomy and renal function decline over 4.7 years.⁹⁶

LUPUS NEPHRITIS

Lupus nephritis is common in Asia (see also Chapter 31). Previous studies showed that systemic lupus erythematosus (SLE) and renal involvement are more common in Asian populations compared with Caucasians.^{97,98} Data from renal biopsy registries showed that lupus nephritis is the leading cause of secondary glomerular diseases, accounting for $>50\%$ of cases in China, Korea, and Japan.^{70,71,99-101}

A meta-analysis of GWAS data showed substantial commonality in shared risk variants between European and Chinese patients with SLE, with overall higher risk variant frequencies in the Chinese, suggesting a greater SLE genetic risk burden in East Asia.¹⁰² Compared with other race or ethnicity, Asian patients may also develop more severe nephritis, resulting in lupus nephritis being an important cause of ESKD in Asia.¹⁰³ Genetic association studies have identified >50 polymorphisms associated with lupus nephritis susceptibility or pathogenesis.^{104,105}

Clinical manifestation and the diagnosis of lupus nephritis in Asian patients are no different from other parts of the world.¹⁰⁶ However, within Asia there is marked variation in the timing of presentation due to variations in health care systems. This has important implications on treatment and patient outcomes because earlier presentation is in

general associated with less irreversible damage. High baseline serum creatinine and significant chronic damage shown in the kidney biopsy portend an unfavorable long-term renal prognosis, whereas histologic features such as vascular or endothelial abnormalities and crescents signify aggressive disease.¹⁰⁷⁻¹¹¹ Studies from China have implicated antineutrophil cytoplasm antibodies in the pathogenesis of crescentic lupus nephritis,^{107,112,113} while a study in Korea reported that anti-Sm antibody was detected in 48.8% of patients with biopsy-proven lupus nephritis and was associated with worse renal prognosis.¹¹⁴ Data from Korea showed that patients presenting after the age of 50 years showed higher renal chronicity features and had inferior outcomes compared with those with disease onset at a younger age.¹¹⁵

Immunosuppressive treatment for lupus nephritis has evolved considerably since the 1980s.^{116,117} Mycophenolate, and more recently tacrolimus, are effective “repurposed” therapies first tested in Asian patients. The shift from corticosteroids combined with cyclophosphamide, followed by azathioprine, to corticosteroids combined with mycophenolate and reduced corticosteroid exposure result in improved clinical outcomes and reduced treatment-associated adverse effects.¹¹⁸ Initial results demonstrating efficacy and safety of mycophenolate therapy in Chinese patients were subsequently confirmed in controlled trials in the United States, Malaysia, and the industry-led international multicenter Aspreva Lupus Management Study.¹¹⁹⁻¹²⁴ Reduced adverse events with mycophenolate treatment compared with cyclophosphamide were associated with improved quality of life.¹²⁵ Results from these trials showed racial variation in response to treatment, with inferior efficacy rates observed in patients of African descent and in Latin America, while high response rates were observed in Chinese patients.^{119,120,126} The high short-term response rates in Chinese patients translated into relatively favorable long-term patient and renal survival and are associated with a relatively low rate of nephritic flares.¹²⁷⁻¹²⁹

Mycophenolic acid pharmacokinetics vary markedly between patients.¹³⁰⁻¹³² Pharmacokinetics data from Thailand showed that the commonly adopted daily mycophenolate mofetil dose of 1.5 to 2 g was associated with a therapeutic level of mycophenolic acid.¹³³⁻¹³⁵ Therapeutic drug level monitoring can be helpful in patients who show unsatisfactory treatment response or drug toxicity, as well as in pediatric patients.¹³⁶

Calcineurin inhibitors suppress cellular immunity and exert a modulatory effect on the podocyte cytoskeleton.¹³⁷⁻¹⁴⁴ Investigators in China reported that a “multitarget therapy” with triple immunosuppression comprising corticosteroids, tacrolimus (4 mg/day), and reduced-dose (1.0 g/day) mycophenolate mofetil was associated with a higher short-term complete remission rate (65%) compared with corticosteroids and intravenous (IV) cyclophosphamide (15%) in patients with combined class IV and class V lupus nephritis¹⁴⁵ and a higher complete remission rate (45.9%) compared with IV cyclophosphamide (25.6%) at 24 weeks in patients with class III/IV/V lupus nephritis,¹⁴⁶ while the renal flare rate (5.47%) was similar to that in controls who received azathioprine (7.62%) as maintenance therapy for 18 months.¹⁴⁷ Investigators in Japan reported a pilot study in 15 patients with lupus nephritis treated with another “multitarget” immunosuppressive treatment regimen that

included prednisolone, IV cyclophosphamide for 3 months, and tacrolimus at a standard dose of 3 mg daily. They reported a complete remission rate of 80% at 6 months, with three patients showing a transient increase of serum creatinine level initially, one cytomegalovirus antigenemia, and one death related to thrombotic microangiopathy.¹⁴⁸ Voclosporin, an analog of cyclosporin previously shown to be noninferior to tacrolimus in the prevention of kidney transplant rejection and with the advantages of being less diabetogenic (compared with tacrolimus) and a more predictable pharmacokinetic-pharmacodynamic relationship, was shown in pivotal and extension clinical trials to be effective in increasing the treatment response rate when added to corticosteroids and mycophenolate, mainly consequent to more effective proteinuria suppression.^{149,150} A report from Hong Kong showed that, instead of adding a calcineurin inhibitor from the start to all patients with active lupus nephritis, the addition of tacrolimus with therapeutic blood level monitoring only in selected patients whose proteinuria had not responded adequately to corticosteroids and mycophenolate was associated with high efficacy and was well tolerated.¹⁵¹ The use of calcineurin inhibitors in patients with lupus nephritis is quite common in Asia, including in stable patients during pregnancy. Extra caution is necessary in patients with significantly impaired kidney function, in view of potential nephrotoxicity.

Belimumab, a monoclonal antibody that inhibits B-cell activating factor (BAFF), has been used in the treatment of SLE for about a decade, showing a satisfactory long-term safety profile. It has demonstrated efficacy in the treatment of lupus nephritis when added to standard therapy of corticosteroids plus either mycophenolate or low-dose cyclophosphamide followed by azathioprine.^{152,153} Importantly, post hoc analysis suggested that the addition of belimumab could improve kidney outcomes and reduce the risk of flares.¹⁵⁴ Safe for patients who have participated in the clinical trials, the real-world experience with voclosporin and belimumab in the treatment of lupus nephritis in Asia is still limited, and there is significant financial barrier impacting the access to treatment.

In Japan, mizoribine was the first drug approved (in 1990) for the indication of lupus nephritis treatment. The mechanism of action is similar to that of mycophenolic acid, and the drug has been largely used during the maintenance therapy for steroid sparing, with generally acceptable tolerability. A postmarketing surveillance study on long-term mizoribine treatment in 559 patients with lupus nephritis showed that overall there was a continued decline in disease activity and 26.5% of patients were in complete remission at 24 months, yet progressive deterioration of kidney function was observed after 12 months' follow-up. Serious adverse drug reactions occurred in 3.6% of patients. Nearly all patients were on concomitant corticosteroid treatment, whereas 51.2% of patients were also receiving tacrolimus.¹⁵⁵

DIABETIC NEPHROPATHY

Diabetic nephropathy (also see [Chapter 41](#)), mostly due to type 2 diabetes in Asia, has become a major cause of ESKD in many Asian cities, in parallel with economic development. In Japan, diabetic nephropathy has overtaken

glomerulonephritis as the leading underlying cause of ESKD in dialysis patients since 1997, and the proportion continues to increase (Fig. 78.3).⁶⁰ In China, diabetic nephropathy is the leading cause of CKD surpassing glomerulonephritis among hospitalized patients.¹⁵⁶ Projection based on Bayesian age-period-cohort analysis indicated that there would be 88,803 deaths from diabetic kidney disease in China in 2030, a 224% increase compared with 1990.¹⁵⁷ The association between diabetes mellitus and CKD has similarly been demonstrated in Korea and Singapore.^{158,159} In a cross-sectional survey that included 1861 patients with type 2 diabetes in Singapore, 53% had concomitant CKD, many with concomitant retinopathy, neuropathy, and cardiovascular disease.¹⁵⁹ Another cross-sectional study that included 2385 Japanese patients showed that 52% had urine albumin-to-creatinine ratio ≥ 30 mg/gCr and/or EGFR < 60 mL/min/1.73 m² and 14% of patients demonstrated an early decline of eGFR.¹⁶⁰

Around 30% to 40% of type 2 diabetic patients develop diabetic nephropathy, the pathogenesis of which involves genetic and environmental factors.^{161,162} Data to date show that genes related to the RAAS, acetyl coenzyme A carboxylase β (ACACB), interleukin-10 (IL-10), interleukin-6 (IL-6), and intercellular adhesion molecule-1 (ICAM1) are involved in the progression of diabetic nephropathy.¹⁶³⁻¹⁷¹ A study in Singapore that included 1950 Asians with type 2 diabetes reported that a higher plasma level of vascular cell adhesion molecule-1 (VCAM-1) was associated with worse renal function and proteinuria.¹⁷² Higher levels of lipoprotein(a) have been associated with the development of CKD in Korean patients with type 2 diabetes,¹⁷³ and data from Chinese patients with type 2 diabetes suggested an association between diabetic nephropathy and apolipoprotein E isoforms, with the ApoE

$\epsilon 2$ allele being more prevalent and the ApoE $\epsilon 4$ allele less prevalent in diabetic patients with nephropathy.¹⁷⁴ Dietary salt intake and serum uric acid level have been reported as independent risk factors, with odds ratios of 1.15 and 2.00, respectively, associated with renal impairment in Japanese patients with type 2 diabetes.¹⁷⁵ Examples of other genetic polymorphisms implicated in susceptibility or progression of diabetic nephropathy include those of ELMO1 in Chinese,¹⁷⁶ 5,10-methenyltetrahydrofolate synthetase in Taiwan,¹⁷⁷ and 2184AG polymorphism in the gene for the receptor for advanced glycation end product in China.¹⁷⁸ In the field of epigenetics, data from Taiwan showed higher levels of miR-21, miR-29a, and miR-192 in diabetic patients with heavier proteinuria and an association among miR-21, miR-29a, miR-29b, and miR-29c with progressive renal failure.¹⁷⁹ Epigenome-wide association studies and multisite analysis have identified CpG sites that are associated with eGFR level and its decline over time, highlighting the significance of methylation markers in CKD risk stratification.¹⁸⁰

The use of sodium-glucose cotransporter 2 inhibitors is rapidly gaining popularity,¹⁸¹ while cardiorenal outcome benefits have also been reported with glucagon-like peptide 1 receptor agonists and nonsteroidal mineralocorticoid receptor antagonists.^{182,183} As in other parts of the world, Asian patients with diabetic nephropathy also show markedly increased cardiovascular risk.¹⁸⁴ In a study that included 563 patients with diabetic nephropathy from Japan and Hong Kong, with 73.5% already receiving treatment with an ACE inhibitor, treatment with the angiotensin receptor blocker olmesartan was shown to reduce proteinuria independent of ACE inhibitor and also reduce adverse cardiovascular outcomes (HR: 0.65, $P = .042$).¹⁸⁵ Post hoc analysis of this trial investigated the impact of systolic blood pressure on renal and cardiovascular outcomes.¹⁸⁶ Patients were followed for a mean of 3.2 years, and the results showed that in patients with proteinuria ≥ 1 g/gCr, a mean follow-up systolic blood pressure of > 130 mm Hg was associated with an HR of 2.33 (1.62–3.36) for renal composite outcome (doubling of serum creatinine, ESKD, or death), compared with patients with follow-up systolic blood pressure ≤ 130 mm Hg. In patients without a history of cardiovascular disease, a follow-up mean systolic blood pressure of > 140 mm Hg was associated with an HR of 2.04 (1.23–3.40) for cardiovascular outcomes compared with those with systolic blood pressure < 140 mm Hg. The median slopes of eGFR were -3.27 , -4.53 , and -7.13 dL/mg/year in patients with systolic blood pressure ≤ 130 , 131–140, and > 140 mm Hg, respectively ($P = .008$ between ≤ 130 and 131–140, $P < .001$ between ≤ 130 and > 140 mm Hg). The investigators thus concluded that a systolic blood pressure ≤ 130 mm Hg was associated with renoprotection.¹⁸⁶ Results from a study in Japan showed limited impact of intensified multifactorial intervention in patients with abnormal serum creatinine on adverse kidney outcomes, highlighting the importance of early intervention.¹⁸⁷

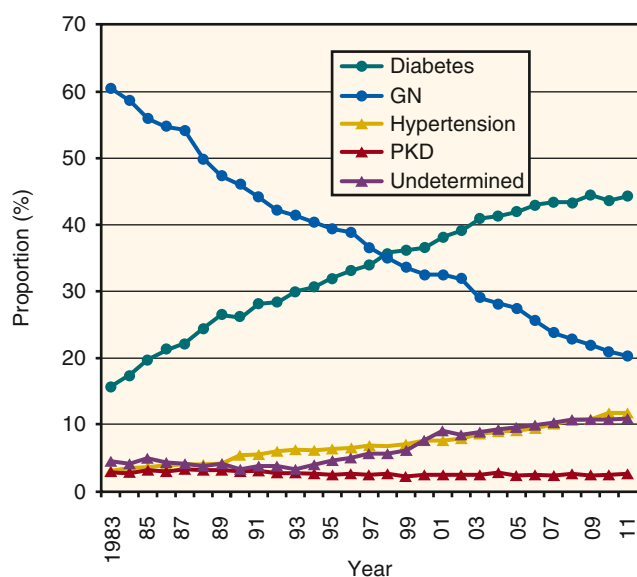


Fig. 78.3 Proportion of primary diagnoses of end-stage renal disease in prevalent patients on dialysis in Japan. GN, Glomerulonephritis; PKD, polycystic kidney disease. (Reprinted with permission from Elsevier. From Hanafusa N, Nakai S, Iseki K, et al. Japanese Society for Dialysis Therapy Renal Data Registry – a window through which we can view the details of Japanese dialysis population. *Kidney Int Suppl.* 2015;5:15–22.)

ARISTOLOCHIC ACID NEPHROPATHY

The story of AA nephropathy began in the early 1990s when a group of young females in Belgium was noted to develop tubulointerstitial nephritis with rapidly progressive deterioration of kidney function following the intake of

slimming medications containing herbal ingredients.^{8,188} Subsequently, the causative nephrotoxic agent was identified as AA,¹⁸⁹ present in the Chinese herb Guang Fang Ji, and poisoning occurred when this was mistaken as Han Fang Ji, a diuretic herb in traditional Chinese medicine.¹⁹⁰ It was also noted that AA exposure predisposed to the development of urothelial malignancies, so AA was classified as a human carcinogen class I by the World Health Organization (WHO) International Agency for Research on Cancer in 2002.^{18,191,192} Kidney disease related to AA exposure is termed “AA nephropathy” (AAN). In the following years, cases of AAN have been reported in other parts of the world, and areas with high prevalence rates include China, Taiwan, and countries in Eastern and Southeastern Europe near the Danube tributaries, where the entity of Balkan endemic nephropathy was recognized since the 1950s.^{193,194} After noting the similarities in renal histopathology and the high incidence of upper tract urothelial cancer between AAN and Balkan endemic nephropathy, the etiologic link to environmental exposure to AA was confirmed.^{195,196} AA released from decaying *Aristolochia clematitis* weed had contaminated soil and food crops. In both conditions, AA-DNA adducts and the hallmark A:T to T:A base transversion are detected in renal cortical and urothelial malignant tissues.¹⁹⁷

AA is present in plants of the *Aristolochia* genus containing nitrophenanthrene carboxylic acids, of which 8-methoxy-6-nitro-phenanthro-(3,4-d)-1,3-dioxolo-5-carboxylic acid (aristolochic acid I, AAI) and 6-nitro-phenanthro-(3,4-d)-1,3-dioxolo-5-carboxylic acid (AAII) are the predominant components.¹⁹⁸ In Asia, exposure to AA is mostly related to intake of herbal remedies, whereas in patients with Balkan endemic nephropathy, AA might have been ingested via bread made with flour contaminated by *A. clematitis*.¹⁹⁹⁻²⁰¹ AA has been identified in different Chinese medicinal herbs (Table 78.2).²⁰⁰ Pathogenesis involves injury to renal tubular cells with altered oxidative pathways and mitochondrial damage, followed by apoptosis and impaired regenerative potential, resulting in interstitial fibrosis and tubular atrophy.²⁰²⁻²⁰⁴ Human urothelium is rich in peroxidases; the aristolactams activated by peroxidase then form adducts with DNA, which have been associated with A:T to T:A base transversion in the *TP53* tumor suppression gene.^{198,201,205,206} *TP53* is involved in cell-cycle checkpoints,

DNA repair, and apoptosis,²⁰⁷ thereby explaining the carcinogenic effect of AA. The causative link between AA exposure and urothelial malignancies emerged from clinical and animal studies.^{17,18,205,208-212} Characteristic AA-DNA adducts in the renal tissue such as 7-(deoxyadenosin-N6-yl) aristolactam I (dA-AAI) can serve as biomarkers of prior AA exposure.^{18,198,201,213} This deoxyadenosine adduct causes characteristic A-T transversion mutations, and such mutations were found in the *TP53* tumor suppressor gene in tumors from AAN and patients with Balkan endemic nephropathy.^{195,214} More recently, using whole-genome and exome sequencing analysis, A:T transversion mutations were also observed in other gene loci.²¹⁵⁻²¹⁷

AAN is characterized by extensive cortical interstitial fibrosis with relative sparing of glomeruli and urinary tract urothelial carcinomas. Most cases presented with kidney failure, moderate hypertension, and severe anemia, and approximately 70% showed rapid progression to ESKD.^{202,218} Some patients presented with Fanconi syndrome.^{219,220} Proteinuria was relatively mild due to its tubular origin.²²¹ Kidney biopsies showed extensive interstitial fibrosis and tubular atrophy, often starting from the peripheral cortex.^{8,188} Interstitial infiltrates were mostly not prominent, though inflammatory infiltration was observed in some renal biopsies, suggesting that immune-mediated injury could not be totally excluded.¹⁹⁴ While there is no pathognomonic diagnostic feature, the pattern of injury in the absence of other diseases would suggest the diagnosis of AAN. In a series of 300 Chinese patients with AAN, 13 patients presented with AKI, 7 patients presented with tubular dysfunction with relatively normal serum creatinine, and 280 patients presented with CKD.²²² AKI cases were associated with high AA exposure and progressed to kidney failure over 1 to 7 years, while patients presenting with predominantly tubular dysfunction and relatively normal serum creatinine showed slow CKD progression rates, suggesting a dose-dependent relationship between AA exposure and the severity of renal injury.²²³ Cumulative exposure to 61–100 g of Mu Tong or Fangchi (herbs that contain AA) had been associated with increased risk of ESKD with odds ratios of 1.47 and 1.60, respectively, increasing to 5.82 and 1.94, respectively, when the exposure was >200 g.²²⁴

Some patients first developed upper urinary tract urothelial carcinoma followed by subsequent bladder cancer, and malignancies may develop at variable periods after the diagnosis of AAN, when patients are already on long-term dialysis or after kidney transplantation.^{18,225} The risk for malignancies is related to the cumulative dose of AA exposure, and cumulative exposure to >200 g of Guang Fang Ji was associated with increased risk of urothelial carcinoma.¹⁸ The incidence of upper tract urothelial cancer appears particularly high in Eastern European patients diagnosed with Balkan endemic nephropathy and in patients with AAN in Taiwan, especially in females.^{226,227}

Ultra-high-performance liquid chromatography–multistage fragmentation mass spectrometry can be used to detect AAI and AAI in herbal products,²²⁸ and hollow fiber liquid-phase microextraction has been used for the extraction and quantitation of AAI in human urine.²²⁹ AA-DNA adducts in the renal cortex or urothelial tissues can serve as biomarkers of prior AA exposure,^{18,198,213} using ³²P labeling or liquid chromatography–mass spectrometry or ultraperformance

Table 78.2 Chinese Medicinal Herbs that Contain Aristolochic Acids

Name of Chinese Herb	Botanical Name
Guang Fang Ji (Fangchi)	<i>Aristolochia fangchi</i>
Xixin	Radix et Rhizoma Asari
Guan Mu Tong	<i>Aristolochia manshuriensis</i>
Qing Mu Xiang	<i>Aristolochia cucurbitifolia</i>
Ma Dou Ling	<i>Aristolochia debilis</i>
Tian Xian Teng	<i>Aristolochia contorta</i>

Modified from Yang HY, Chen PC, Wang JD. Chinese herbs containing aristolochic acid associated with renal failure and urothelial carcinoma: a review from epidemiologic observations to causal inference. *Biomed Res Int*. 2014;2014:569325.

liquid chromatography–triple quadruple mass spectrometry of exfoliated urothelial cells.²³⁰⁻²³³

Continued exposure to AA should be stopped. Standard algorithms for diagnosis and management have been devised for subjects at increased risk of Balkan endemic nephropathy based on geographic location or family history.¹⁹³ In view of the inflammatory cell infiltration in the renal interstitium,²³⁴ the effect of corticosteroid therapy has been investigated. Data on Caucasian AAN patients suggest that corticosteroid therapy may reduce the rate of progression to ESKD.^{235,236} In a trial that included 43 Chinese patients with AAN, 25 patients were treated with prednisolone at 0.5 mg/kg body weight daily for 1–3 months, then maintained at 10 mg daily. The results suggested slower rate of CKD progression in the steroid-treated group.²³⁷ Patients exposed to AA should have regular follow-up for the monitoring of kidney function and development of uroepithelial malignancies. Patients with histopathologic features of endemic nephropathy and those in CKD stage 3 or above are at increased risk of developing upper tract urothelial cancer and are recommended to have urine cytology and imaging every 6 months according to the European consensus on endemic nephropathy.¹⁹³ This also applies to patients with previous bladder cancer. In view of the high incidence of urothelial carcinoma in patients with ESKD due to endemic nephropathy, prophylactic bilateral nephroureterectomy has been recommended with determination of the level of AA-DNA adducts in the resected renal cortical tissue, especially in kidney transplant recipients before or after transplantation.^{193,238}

ANTINEUTROPHIL CYTOPLASMIC ANTIBODY–ASSOCIATED VASCULITIDES

Antineutrophil cytoplasmic antibody (ANCA)–associated vasculitis (AAV) is a common autoimmune disease in Caucasians, with an annual incidence rate of 10 to 20 pmp and peak age of disease onset of 65 to 74 years (see also Chapter 32).²³⁹⁻²⁴¹ In Asia, AAV is an important glomerular disease in elderly patients presenting with AKI.²⁴² In contrast to Caucasian populations, where GPA and anti-PR3 are more common, data from China and Japan showed a preponderance of MPA and anti-MPO disease, often associated with more chronic lesions in the kidney biopsy.²⁴³⁻²⁴⁸ MPA or GPA mostly occurs in older adults, although it has been reported at all ages. A series that included 234 Chinese patients with AAV showed that >40% of patients were older than the age of 65 years at diagnosis and 94.9% of the elderly patients had anti-MPO antibodies.²⁴⁹ Data from Japan also showed that the annual incidence of MPA was approximately 10 times higher in subjects older than the age of 65 years than in younger adults.²⁴⁷ In addition, compared with younger patients, older patients had more pulmonary involvement and a higher risk of infections. While the typical presentation is rapidly progressive glomerulonephritis, a retrospective study in China reported that 27 (12.1%) of 223 patients with ANCA-GN showed concomitant membranous nephropathy, associated with a higher level of proteinuria and more severe renal injury and tubulointerstitial disease, and a relatively low seropositive rate for anti-PLA2R antibodies compared with patients with primary membranous nephropathy (12.5% vs. 65.0%).²⁵⁰

Initial treatment of active ANCA-associated vasculitis with kidney involvement is corticosteroids combined

with cyclophosphamide or rituximab, while low-dose corticosteroids combined with either rituximab (preferred) or azathioprine is the recommended maintenance immunosuppressive therapy.^{116,251} Data from a study in China that included 398 patients with AAV showed that 135 (33.9%) died during follow-up, with 83 deaths occurring within the first 12 months and infection being the leading cause of death (39 cases).²⁵² In patients presenting with kidney failure that required dialysis, infection (47.4%) and vasculitis (26.3%) were the leading causes of death.²⁵³ The high incidence and poor outcome of infections have also been reported from Japan.^{254,255} Data from a retrospective study in 82 Japanese patients with ANCA-positive MPA, 29 of whom were treated with corticosteroids plus cyclophosphamide while 53 were treated with corticosteroids alone, showed similar remission rates of >90% in both groups but a lower 5-year survival rate in those given cyclophosphamide (50% vs. 73% in the steroid-only group, $P = .041$).²⁵⁶ Another study reported that disease severity at presentation was associated with both the risk of relapse and infective complications during follow-up.²⁵⁷

Clinical outcomes including survival of patients with AAV have improved over the past decade, attributed to earlier diagnosis and improved treatment.^{255,258}

HEPATITIS B/C–ASSOCIATED RENAL DISEASES

Infection with HBV or hepatitis C virus (HCV) may be associated with a variety of glomerular diseases due to immune complex deposition or immune complex–mediated vasculitis,²⁵⁹⁻²⁶² with or without cryoglobulinemia,²⁶³ the latter being more common in the case of HCV infection (see also Chapter 34).

In endemic areas, such as some Asian countries, the transmission of HBV is mostly vertical from infected mother to child, followed by horizontal transmission within families or horizontal infection of adults who do not have protective immunity. In areas of low prevalence, HBV is primarily a disease of adolescents and young adults, through sexual or parenteral acquisition.²⁶⁴ In the Far East before the advent of vaccination programs, vertical maternal–infant transmission of HBV during the perinatal period had been the predominant route of infection accounting for persistent endemicity.²⁶⁵ In addition to specific forms of glomerular diseases, HBV infection is associated with increased incidence and prevalence rates of CKD in endemic regions such as Taiwan, China, and South Korea.²⁶⁶⁻²⁶⁸ Analysis of nationwide health insurance data showed that antiviral treatment with nucleos(t)ide analogs was associated with a reduced risk of developing ESKD in CKD patients with HBV infection, with 16-year cumulative rates of 10.1% and 2.2% in the untreated and antiviral-treated populations, respectively ($P < 0.0001$).²⁶⁹

The age of presentation for HBV-associated kidney diseases thus varies according to the time of infection. Mass HBV vaccination programs markedly reduced the HBsAg carriage rate and consequent HBV-associated liver complications including hepatocellular carcinoma.²⁷⁰ For example, the nationwide HBV vaccination program of Taiwan that was introduced in 1984 reduced the HBsAg carrier rate in children from 9.8% to <1% in 2 decades.²⁷¹ The frequency of HBV-associated membranous nephropathy among children infected

with HBV also decreased from 11.6% in the period 1974–1984 to 2.1% in 1994–2004 and to 0% in 2004–2009.²⁷² A similar impact of immunization program on HBV-associated membranous nephropathy, reducing the incidence by approximately 90%, had previously been observed in South Africa.²⁷³

In a report from China that included 5935 kidney biopsies performed in 2010–2015, of which 1038 (17.5%) were diagnosed as secondary glomerular diseases, HBV-associated glomerulonephritis accounted for 9.2% of the secondary cases.²⁷⁴ The most common pattern of renal involvement in renal biopsy is membranous nephropathy, followed by membranoproliferative glomerulonephritis, polyarteritis nodosa, IgAN, and focal segmental glomerulosclerosis, although the latter two categories might represent coincidental findings.²⁶⁰ HBV-associated membranous nephropathy is more common in children and in many cases resolves spontaneously when seroconversion from HBeAg seropositivity to anti-HBe seropositive status occurs, but spontaneous resolution of HBV-related glomerulonephritis is uncommon in adults and some patients show progressive renal deterioration over time.²⁷⁵ Confirmation of HBV etiology is by the demonstration of HBV antigens in the kidney biopsy, although this may not be positive in all cases and positive staining needs to be interpreted in the context of appropriate histopathologic features.²⁷⁶

Most data on the treatment of HBV-associated glomerular diseases are derived from patients with membranous nephropathy. Treatment with interferon or nucleoside analogs has been reported to lead to a reduction of proteinuria in patients with HBV-related membranous nephropathy. Interferon treatment given for 4 to 12 months was associated with sustained remission of proteinuria in 20% to 100% of patients, usually occurring within 6 months of clearance of HBeAg.²⁷⁵

The M-type phospholipase A2 receptor (PLA2R) on the surface of podocytes has been identified as the major autoantigen in primary membranous nephropathy, and autoantibodies directed against PLA2R are mostly of IgG4 subclass.^{277,278} Up to 80% of patients with primary membranous nephropathy, including patients in Asia, have circulating anti-PLA2R antibodies.^{279–284} Data from a study in China that included 179 patients with primary membranous nephropathy, 40 patients with membranous lupus nephritis, and 26 patients with HBV-associated membranous nephropathy showed glomerular PLA2R expression, in a fine granular pattern along the glomerular capillary wall, in 92.2% of patients with primary membranous nephropathy and 7.7% of patients with HBV-associated membranous nephropathy.²⁸⁵ Also, 93.3% of the primary membranous nephropathy cases showed prominent glomerular IgG4 deposition, while IgG3 was the predominant subclass in patients with HBV-associated membranous nephropathy, with IgG4 present in 11.5%.²⁸⁵

While more often associated with HCV than HBV,^{263,286–288} cryoglobulinemic renal disease has been reported in China, which has an HBV prevalence rate of 7.18% in the general population.²⁸⁹ A series of 12 patients diagnosed with HBV-associated cryoglobulinemia in 2008–2015 showed that renal involvement, characterized by membranoproliferative glomerulonephritis and nephrotic-range proteinuria with microscopic hematuria, affected 75% of patients while 58.3%

presented with skin rash.²⁹⁰ All patients showed reduced complement level and tested positive for rheumatoid factor.²⁹⁰

The data to date suggest that antiviral therapy improves HBV-associated membranous nephropathy. However, this conclusion is based on a relatively small number of subjects and not from randomized controlled trials.^{291–293} A meta-analysis that included 325 Asian patients with HBV-associated glomerulonephritis showed an overall efficacy of nucleoside/tide in reducing proteinuria, resulting in a 3.6-fold increased rate of proteinuria remission.²⁹⁴ Another meta-analysis that included 127 patients also yielded results supportive of treatment efficacy, applicable to both interferon and nucleoside/tide analogs.²⁹⁵

Antiviral therapy with entecavir is generally recommended in patients with HBV-associated renal disease and evidence of active viral replication as indicated by the level of HBV DNA.^{296–299} In patients with HBV-associated renal disease, immunosuppression is reserved for those with rapidly progressive glomerulonephritis or severe polyarteritis nodosa, with plasmapheresis considered in cryoglobulinemic cases.

Recent epidemiologic reports showed a prevalence rate of approximately 3% to 4% for HCV infection in the general population across Asia, varying from 0.08% in Hong Kong, 0.37% in Singapore, 0.47% in the Philippines, 0.49% in Japan, 0.6%–1.3% in Korea, 1.6%–3.2% in China, 2.2% in Thailand, to 1.8%–4.4% in Taiwan.^{300–303} The predominant genotypes also vary, with 1b accounting for around 70% in China and around 40% in Korea, whereas genotype 3 accounts for up to 80% of cases in India and Pakistan, and genotype 6 restricted to a few places such as Hong Kong and Thailand.³⁰¹ Prevalence is higher with increasing age, among injection drug users (48%–90%), and in those with coinfection by human immunodeficiency virus (32%–85%). Annual incidence of HCV in China was estimated as 6.01/100,000.³⁰²

Extrahepatic manifestations occur in 38% to 76% of patients with chronic HCV infection, mostly as immune-mediated diseases such as mixed cryoglobulinemia, Sjögren syndrome, autoimmune thyroiditis, and B-cell lymphoproliferative disorders.³⁰⁴ Renal involvement, mostly as membranoproliferative glomerulonephritis, occurs in approximately 20% of patients with mixed cryoglobulinemia.³⁰⁵ For extrahepatic manifestations of HCV, treatment with direct-acting antiviral agents has demonstrated efficacy and is recommended.^{305,306} Combination therapies such as glecaprevir/pibrentasvir or sofosbuvir/velpatasvir are commonly used since they cover HCV genotypes 1 to 6. Antiviral efficacy and improvement of renal abnormalities have been demonstrated in Asian patients with HCV-associated cryoglobulinemic membranoproliferative glomerulonephritis.^{307,308} However, recurrent mixed cryoglobulinemia may occur despite sustained virologic response after treatment.³⁰⁹ In addition, the high cost of direct-acting antiviral agents presents a substantial financial barrier to treatment access in many Asian countries, as is the case with rituximab for the treatment of HCV-associated cryoglobulinemic vasculitis.

GENETIC KIDNEY DISEASES

Autosomal dominant polycystic kidney disease accounts for approximately 5% of ESKD in adults (see also [Chapter 44](#)). Post hoc analysis results from the TEMPO 3:4 and TEMPO

Extension trials in Japan showed beneficial effect of tolvaptan in reducing eGFR decline irrespective of kidney volume change and sustained efficacy with prolonged tolvaptan treatment.^{310,311} A postmarketing surveillance study showed efficacy of tolvaptan treatment on kidney volume and eGFR decline that was similar to that reported previously in pivotal trials, with abnormal liver biochemistry noted in 8.3% of patients.³¹²

Alport syndrome is a well-recognized cause of kidney failure, due to inherited genetic variants that result in altered structure of type IV collagen. Novel *COL4A5* variants associated with Alport syndrome has been reported,³¹³ and studies in Chinese and Japanese patients with x-linked Alport syndrome showed that truncating mutations in the *COL4A5* gene was associated with inferior kidney survival, as well as genotype-phenotype association with the impact of renin-angiotensin blockade on progressive CKD.³¹⁴⁻³¹⁶

The histologic lesion termed “focal segmental glomerulosclerosis” (FSGS) can be secondary to heterogenous kidney injuries or can result from genetic variants that affect podocyte-related or collagen IV-related genes.³¹⁷ A missense homozygous mutation in the mitochondrially encoded tRNA leucine 1 cosegregating with a *COL4A4* missense mutation was identified as the cause of familial hematuria and FSGS in a Chinese family.³¹⁸ In a study that included 35 adult and 18 pediatric Thai patients with biopsy-proven focal segmental glomerulosclerosis, whole-exome sequencing revealed disease-associated pathogenic or likely pathogenic genetic variants in six patients (11.3% overall; 11.4% among adults, 11.1% among pediatric patients), with three patients having a disease-associated variant in the *COL4A4* gene.³¹⁹ There has also been progress in the genetic diagnosis of nephrophtosis, an important cause of kidney failure in children and adolescents,³²⁰ and autosomal dominant tubulointerstitial disease.³²¹ Also, genome-wide associated studies in Japan and Taiwan have identified susceptibility gene loci that were associated with CKD and/or proteinuria.^{322,323}

KIDNEY DISEASES RELATED TO INFECTIONS

For more information about kidney injury related to infections, please also see [Chapters 34, 59, 75, 76, and 77](#).

LEPTOSPIROSIS

Leptospirosis is the most widespread zoonosis and is endemic worldwide, being most prevalent in tropical, humid, and rural environments, particularly in warm and humid regions.³²⁴ The disease is caused by the spirochete *Leptospira*, which enters the human body through direct cutaneous or mucosal transmission or by aerosolization.³²⁵ Rodents and small mammals are reservoirs for *Leptospira*, and infected animals shed the spirochetes in their urine into water or soil, which then infect humans through occupational (as occurs in farmers, sewage workers, and miners), household environmental, or recreational exposure.

Approximately 90% of *Leptospira* infections are subclinical or associated with mild self-limited disease. Leptospirosis is an important topic in travel medicine because the disease can present as an imported case in Western countries in travelers returning from Asia.³²⁶ However, it can also

occur in subjects with no recent travel history and in urban settings,³²⁷ especially in areas with a large population of rodents. Leptospirosis occurring in an endemic setting or presenting as epidemic outbreaks, especially in developing areas and crowded urban areas, occurs in different parts of the world including Thailand, the Philippines, and China.³²⁸⁻³³¹ It has been recognized as a reemerging infectious disease in Japan,³³² and a high prevalence rate of up to 50% of pathogenic *Leptospira* has been detected in wild and domesticated animals including rats and dogs in an endemic area in the Jiangxi Province of China.³³³

Symptoms develop after an incubation period of 5 to 14 days. Mild disease has nonspecific symptoms that are indistinguishable from other febrile illnesses. Severe and potentially fatal disease characterized by fever; headache; myalgia; AKI due to acute interstitial nephritis, jaundice, and hemorrhagic complications; and multiorgan failure occurs in up to 10% of patients and is associated with a mortality rate of 5% to 15%.³²⁵ Typical clinical manifestations of leptospirosis fall into four categories: 1. mild influenza-like illness; 2. Weil syndrome characterized by jaundice, renal failure, hemorrhage, and myocarditis with arrhythmias; 3. meningitis/meningoencephalitis; and 4. pulmonary hemorrhage with respiratory failure.³³⁴ Infection with icterohemorrhagiae serogroups has been reported to be associated with an increased risk of severe disease or death.^{324,335} Cardinal features of severe leptospirosis include nonoliguric AKI, marked hyperbilirubinemia of >30 mg/dL (512 μmol/L) with less severe elevation of aminotransferase levels, thrombocytopenia, and lung involvement.³²⁴ AKI as a result of acute tubulointerstitial nephritis is an early manifestation.³³⁶ Oliguria, cardiac arrhythmia, and pulmonary involvement portend a poor prognosis.³³⁷ Urinary sodium and potassium loss due to proximal renal tubular dysfunction resulting in electrolyte abnormalities is common.³³⁸ Proximal tubular dysfunction such as bicarbonaturia, glycosuria, decreased proximal sodium reabsorption, phosphaturia, magnesuria, and uricosuria can occur.³³⁹ Rhabdomyolysis can also contribute to the AKI.³⁴⁰ The manifestation of severe disease can be variable. In some patients after the initial septicemic phase, there is a period of transient symptomatic improvement before an immune phase in which critical disease manifestations can occur, whereas in other patients there is progressive symptomatic deterioration to fulminant disease.³²⁴ The triad of fever, jaundice, and AKI in an acutely ill patient should alert the clinician of possible leptospirosis.

Leptospira reach the kidneys via hematogenous spread and finally localize in the proximal tubular lumen.³⁴¹ Data from animal studies showed renal tubular cell injury, acute and chronic interstitial nephritis, and associated microvascular injury, particularly at the corticomedullary junction.^{342,343} *Leptospira* outer membrane proteins (OMPs) may elicit tubular injury and inflammation through Toll-like receptor (TLR)-dependent pathways.³³⁶ LipL32, a highly conserved leptospiral lipoprotein, induces inflammatory cytokines and stimulates extracellular matrix production in cultured renal tubular epithelial cells. It adheres to components of extracellular matrix,³⁴⁴⁻³⁴⁷ mediated through Toll-like receptor 2 (TLR2).³⁴⁸⁻³⁵⁰ In chronic infection, *Leptospira* may colonize and persist in the proximal renal tubule, and this has been associated with chronic tubulointerstitial nephritis and fibrosis.³⁵¹ Results from an epidemiologic study

suggested that previous or chronic leptospiral infection is a risk factor for CKD.³⁵²

The diagnosis of leptospirosis is by serologic methods such as the microscopic agglutination test to detect antibody to *Leptospira*, polymerase chain reaction (PCR) to detect *Leptospira* DNA, or rarely culture of the microorganism, which requires specialized culture media and takes weeks.^{324,325} Serologic testing is often negative early in the course of the disease,³⁵³ and suboptimal sensitivity and specificity rates have been reported with an IgM enzyme-linked immunosorbent assay (ELISA) for the diagnosis of acute leptospirosis.³⁵⁴ PCR-based nucleic acid amplification testing of blood, urine, or cerebrospinal fluid is much more sensitive and facilitates early diagnosis.³²⁷

Prompt treatment with antibiotics in suspected cases before microbiologic confirmation and in confirmed cases is recommended.^{334,337,355,356} Serum creatinine level improved after 2.7 ± 2.2 days of IV penicillin therapy.³⁵⁷ Delayed initiation of antibiotics by ≥ 2 days was associated with more severe disease.³³⁵ IV penicillin is used to treat severe disease, and oral doxycycline is used to treat mild disease.³⁵⁸ Adults with severe disease should receive either penicillin (1.5 million units IV every 6 hours), doxycycline (100 mg IV twice daily), ceftriaxone (1–2 g IV once daily), or cefotaxime (1 g IV every 6 hours) for a duration of at least 7 days. Pregnant women with severe leptospirosis may be treated with IV penicillin, ceftriaxone, cefotaxime, or azithromycin but not doxycycline. Patients with mild disease can be treated with doxycycline (adults: 100 mg orally twice daily) or azithromycin (adults: 500 mg orally once daily for 3 days), and pregnant women should be treated with either azithromycin or amoxicillin. Jarisch-Herxheimer reactions may occur following effective treatment. It is advisable to follow the kidney function of patients in view of the association with CKD.³⁵²

DENGUE VIRAL INFECTION

Dengue is a common arthropod-borne viral (arbovirus) disease in tropical and subtropical regions.^{359,360,361} While the annual incidence is up to 400 million by the WHO,^{362,363} this could be an underestimate because the symptoms and signs are nonspecific, with fever being one of the most common clinical manifestations.³⁶⁴ It is endemic in >100 countries, and an estimated 2.5 billion of the world's population are at risk of dengue infection.^{365,366} The incidence of dengue fever has increased 30-fold in the past 50 years, and factors such as global warming, increased air travel and globalization, urbanization, and ineffective mosquito control have resulted in the spread of dengue virus and the introduction of multiple viral serotypes into permissive areas, resulting in most tropical regions becoming hyperendemic with multiple viral serotypes cocirculating.³⁶⁷⁻³⁷³ The disease is common in South and Southeast Asia in countries such as Malaysia, Thailand, Indonesia, Vietnam, and Singapore^{359,374-376} and is spreading to other parts of Asia such as Japan, Korea, China, Taiwan, and Hong Kong.^{364,368-372,377}

Dengue is transmitted by *Aedes* mosquitoes, principally *Aedes aegypti*, and is caused by four viral serotypes of the genus *Flavivirus* (DENV-1 to DENV-4).³⁶⁵ Infection with one serotype confers lifelong immunity to that type but only short-term immunity to the others, and subsequent infection

with a different serotype is associated with increased risk of severe complications. An infected mosquito is infective for life. There is no specific antimicrobial therapy. Prevention is by reducing the habitat and number of mosquitoes and limiting exposure to bites. Dengvaxia is a tetravalent live-attenuated dengue vaccine approved by the U.S. Food and Drug Administration for use in individuals aged 9 to 16 years with laboratory-confirmed previous dengue infection and living in endemic areas.^{363,378} Dengvaxia may increase the risk of severe dengue in those with no prior history of infection, due to antibody-dependent enhancement (see later). Recently, the tetravalent live-attenuated dengue virus vaccine Qdenga (previously known as “TAK-003”) has been approved in the European Union and Brazil for use in subjects who are 4 years of age or older and in Indonesia for those aged 6 to 45 years.³⁷⁹

The time from infection to onset of illness, termed the “intrinsic incubation period in humans,” ranges from 3 to 14 days, with an average of 4 to 7 days. Symptoms may include fever (often the first manifestation and can be the only manifestation in children), headache, myalgia, joint pain, and a measles-like skin rash, which lasts up to 7 days. Slight abdominal pain and diarrhea can occur. Leukocyte counts are usually decreased. Patients can become better or deteriorate after 2 to 5 days. A small proportion of patients develop life-threatening dengue hemorrhagic fever (DHF), with features of thrombocytopenia, petechiae or ecchymosis, disseminated intravascular coagulation, and plasma leak resulting in hemoconcentration, hypoproteinemia, pleural effusion, and ascites. More severe cases are termed dengue shock syndrome (DSS), in which systemic vascular leak and severe hypotension occur.³⁶⁴ There can be concomitant hepatitis, neurologic disorders, and myocarditis. Mortality rate in untreated cases could reach 20%, while supportive treatment including IV hydration can reduce the mortality rate to <1%.³⁶⁵ In 2009, the WHO revised the classification as dengue with or without warning signs, and severe dengue, with the objective of facilitating the identification of patients at risk of severe disease.³⁶⁷ It has been suggested that severe manifestations such as DHF and DSS occur as a result of antibody-dependent enhancement when a dengue-immune subject acquires infection by a different DENV serotype.³⁸⁰ The facilitation of DENV lineage dispersal by air traffic of humans and/or mosquitoes thus has potential clinical and public health implications, and investigations have shown that large transportation hubs such as Thailand could act as central points of DENV lineage movement in Asia.³⁷³

Proteinuria is common in severe dengue, and a detection rate of 74% has been reported.³⁸¹ The underlying pathogenesis of renal abnormalities in dengue is probably multifactorial. AKI is an infrequent complication of dengue infection, occurring in up to 13% of patients depending on the severity of infection, but is associated with increased morbidity and mortality, and a mortality rate of 64% has been reported.³⁸²⁻³⁸⁵ In severe dengue infection, hemodynamic disturbance, acute hemolysis, and rhabdomyolysis contribute to direct kidney injury.³⁸⁶⁻³⁸⁹ Immunologic mechanisms might also play a role, as suggested by renal abnormalities in the absence of shock, the demonstration of immune complex-mediated mesangial proliferation in human and animal models, and association with reduced levels of complement components.^{390,391}

Anti-DENV IgM antibody detection with ELISA is most commonly used for diagnosis of acute disease, while patients in convalescence are identified through IgM and IgG seroconversion.³⁶⁷ DENV viremia is detectable 24 to 48 hours before fever onset and continues for 5 to 6 days. DENV RNA or NS1 detection is useful for early virologic diagnosis. Antibody cross-reaction is observed between DENV and Zika virus because both are Flaviviridae. Specific detection of DENV, chikungunya, and Zika viral RNA can be done by real-time PCR in acute samples.

Supportive care and careful monitoring are essential, as there is no specific treatment for dengue. Fluid intake during the 24 hours before being seen by a clinician has been significantly associated with decreased risk for hospitalization.³⁹² Fluid therapy is key to dengue management and is applied on the basis of disease severity, and fluid lost due to capillary leakage must be replaced with IV crystalloid solutions (volume-expanding fluid replacements such as lactated Ringer solution) or physiologic (normal) saline solution. The subsidence of fever could portend clinical deterioration, and thus patients need to be carefully monitored. Analgesic and antipyretic drugs such as paracetamol can be prescribed at the usual dosage for children and adults, although care should be exercised in patients with hepatitis. Clinical management guidelines advise against the use of aspirin or nonsteroidal antiinflammatory drugs for fear of increasing the bleeding risk. Treatment with corticosteroid or chloroquine was shown to be not beneficial.^{393,394} Vector control to avoid mosquito bites should be used to prevent disease transmission.

MALARIA

Malaria is a protozoan disease caused by four main *Plasmodium* species (*Plasmodium falciparum*, *Plasmodium vivax*, *Plasmodium malariae*, and *Plasmodium ovale*) and the zoonotic parasite *Plasmodium knowlesi* found in many Southeast Asian countries.^{395,396} Approximately 250 million cases of malaria occur annually, with approximately 620,000 deaths.³⁹⁵ The majority of infections occur in Africa (90%), followed by South/Southeast Asia (6%) and the eastern Mediterranean region (2%). Morbidity and mortality due to malaria have decreased globally secondary to malaria control programs over the past few decades, while a high level of transmission continues in some regions. The emergence of artemisinin-resistant *P. falciparum* malaria in Southeast Asia presents a significant therapeutic challenge. Knowledge of local antimalarial drug resistance patterns is important for effective therapy (see [Chapter 59](#)). Effective vaccines such as Mosquirix and R21 have become available.^{397,398}

Malaria is transmitted by *Anopheles* mosquito. Infection in an individual without prior exposure almost invariably produces a febrile illness. The accompanying symptoms are nonspecific and often include rigors, headache, nausea, and muscle pains. If treated with appropriate drugs at this stage, the symptoms remit over a few days, though often with considerable exhaustion. In the case of *P. falciparum*, complete treatment will eradicate the infection, and any return of symptoms reflects incomplete treatment, resistance to the drugs used, or new infection. In the case of *P. vivax* and *P. ovale*, subsequent infections may recur

at intervals as a result of reactivation of the dormant liver-resident hypnozoite stage, unless this is cleared by a prolonged treatment with an 8-aminoquinoline drug.³⁹⁹⁻⁴⁰¹ Aggressive treatment is required, especially for travelers who have no prior immunity.^{400,402} Commonly used methods for laboratory diagnosis include microscopic examination of stained blood films and detection of parasite antigen or nucleic acid,⁴⁰³ of which microscopic examination of thick and thin blood films remains the gold standard. Rapid antigen detection methods and molecular amplification tests are used increasingly. Tests for antiplasmodial antibodies are commercially available but not recommended for diagnosis of acute disease.

About 120 types of *Plasmodium* species have been reported, although only 5 are accepted as human malaria parasites. *P. falciparum* and *P. vivax* cause most of the human infections worldwide. The most virulent species *P. falciparum* is also the most prevalent parasite in Africa, whereas *P. vivax* is the most widely distributed parasite outside of Africa, for example, in India.⁴⁰⁴⁻⁴⁰⁷ A study of 2701 blood samples collected from the China–Myanmar border region using nested PCR to detect the protozoan genome identified 561 malaria cases, including 161 *P. falciparum*, 327 *P. vivax*, 66 *P. falciparum*–*P. vivax* mixed infections, 4 *P. malariae*, and 3 *P. ovale* spp. *P. vivax*, and *P. falciparum* accounted for >60% and approximately 30% of all malaria cases, respectively.⁴⁰⁸ In addition to South Asian countries, *P. malariae* infections have been described in China,⁴⁰⁹ Thailand,^{410,411} and Vietnam⁴¹² and may exhibit high genetic diversity.⁴¹³ Most *P. ovale* spp. reported in China have been imported, mainly from Africa.⁴¹⁴ Indigenous malaria cases decreased significantly after the National Malaria Elimination Action Plan was initiated in China in 2010.⁴¹⁵ A report from Jiangsu province in China showed that of the 1268 cases of malaria reported in the province in 2011–2014, 83.4% were imported *P. falciparum* cases, while 13.0% were due to *P. ovale* and 2.8% were due to *P. malariae*, many of which were imported from countries in Africa and South Asia.⁴¹⁶

In Korea, *P. falciparum* infection was once present among IV drug abusers, while indigenous *P. falciparum* malaria has not been reported since 1945.⁴¹⁷ *P. malariae* was also present before 1945 but has not occurred since. Vivax malaria had been prevalent in Korea for many centuries, but its incidence decreased rapidly from the 1970s. South Korea was once declared as malaria free in the late 1970s, but vivax malaria reemerged in 1993. Initially localized to the area bordering North Korea, it has subsequently spread from west to east.⁴¹⁸⁻⁴²⁰

In the nonendemic setting of Singapore, a retrospective analysis of 214 patients with smear-positive malaria treated at a tertiary hospital in 2000–2010 showed that *P. vivax* accounted for 59.3% of cases, whereas *P. falciparum*, *P. malariae*, and mixed infections accounted for 38.8%, 0.3%, and 1.4% of cases, respectively.⁴²¹ Most patients presented with fever, and thrombocytopenia was common. The 43 severe cases were characterized by older patient age and higher levels of parasitemia. All the patients survived.

A report that included 2058 imported cases of malaria over the period 2011–2019 in Sarawak, Malaysia, showed that 53% were *P. vivax* and 36% were *P. falciparum*, 89% worked in the logging industry, and 46% contracted malaria

in Papua New Guinea.⁴²² *P. knowlesi* has emerged as the most common cause of malaria infection in Malaysia.^{423,424}

Previous reports from India showed that the clinical presentation of severe *P. falciparum* malaria has more complications including multiorgan failure, often with AKI, which was associated with a higher mortality rate.⁴²⁵⁻⁴²⁷ Disease severity is usually lower with vivax malaria compared with *P. falciparum* malaria. However, there are marked variations and there have been increasing reports of severe vivax malaria including respiratory distress,⁴²⁸ AKI,⁴²⁹ and cerebral malaria,⁴³⁰ as well as reports of disease occurring in temperate regions.⁴³¹ In a series that included 341 Korean patients diagnosed in 2005–2009, including 6 recurrent cases, the common manifestations included fever (98.8%), thrombocytopenia (99.7%), chills (62.5%), transaminasemia (54.8%), leukopenia (43.4%), headache (33.7%), anemia (28.5%), and myalgia (24.9%). Less common but severe complications included hypotension (4.1%), altered mentation (0.9%), AKI (0.9%), spleen infarction (0.6%), and spleen rupture (0.3%).⁴³¹ Another series that included 210 Korean patients diagnosed with vivax malaria in 2006–2012 showed that 11 (5.2%) required intensive care unit admission, and among these patients, 5 required mechanical ventilation and 1 was treated with extracorporeal membrane oxygenation.⁴¹⁷ Overall, the disease was categorized as severe in 21.0%, with pulmonary complications being most common (21.9%), followed by cerebral (2.4%), shock (1.9%), and bleeding complications (1.4%), and AKI occurred in 1.0% of patients. None of the patients died.

Pathogenesis leading to organ injury in severe malaria involves a complex interplay between direct damage due to parasite sequestration in microcirculation and host response with the release of various cytokines.⁴³²⁻⁴³⁵ Parasitized red cells adhere to vascular endothelial cell surface markers such as ICAM-1.^{436,437} The incidence of AKI can be up to 60%.⁴³⁸ AKI complicating *P. falciparum* seems to have increased over the past decade.⁴³⁹⁻⁴⁴¹ In addition, AKI has been associated with vivax malaria.^{429,442} AKI complicating *P. falciparum* malaria is associated with acute tubular necrosis. A report from Thailand showed predilection of injury at the distal tubule with features of cell vacuolation and degeneration, necrosis, detachment from the basement membrane, hemosiderin deposition, and minimal chronic inflammatory cell infiltration.⁴⁴³

P. malariae, *P. ovale*, and *P. vivax* may cause febrile attacks years after exposure.⁴⁴⁴ Delayed clinical disease due to *P. ovale* or *P. vivax* is due to the existence of hypnozoites, a minority of parasites that remain in the liver instead of inducing a blood stage like the majority. These dormant microorganisms can remain viable for many years, thereby causing late onset of disease. For *P. malariae*, a low level of parasitemia may persist for years, sometimes below the level of detection on a thick film, and disease can emerge more than a decade after primary infection.⁴⁰⁶ Chronic *P. malariae* infection can result in immune complex formation, leading to membranoproliferative or mesangioproliferative glomerulonephritis and proteinuria (quartan malarial nephropathy).^{445,446}

Owing to its 24-hour erythrocytic stage, *P. knowlesi* replicates quickly.^{447,448} It can cause severe and fatal disease and thus it is important to diagnose and treat early.⁴⁴⁹⁻⁴⁵² *P. knowlesi*

malaria shows a fatality rate that is comparable with, if not higher than, that of *P. falciparum* malaria.^{450,451} The severity of *P. knowlesi* malaria including the complication of AKI is significantly associated with high parasitemia level.^{451,453,454} Diagnosis with blood film microscopy may be associated with misidentification because the *P. knowlesi* ring stages resemble *P. falciparum* and later trophozoites resemble *P. malariae*. Newer diagnostic methodologies such as loop-mediated isothermal amplification are being investigated.⁴⁵⁵

SCRUB TYPHUS

Scrub typhus is a mite-borne infectious disease caused by the intracellular gram-negative bacteria *Orientia tsutsugamushi* (former name *Rickettsia tsutsugamushi*). Rodents are the main reservoir, and their mites act as both the reservoir and vector. Human infection occurs when the larvae of the trombiculid mite infected with *O. tsutsugamushi* bite to suck human tissue fluid.^{456,457} An estimated 1 million cases occur each year, with a high fatality rate of around 30% if not treated early.⁴⁵⁸ It is common in Southeast Asia, Japan, China, and South Korea.^{459,460} Disease manifestations vary from mild or asymptomatic infection to severe potentially fatal illness, with clinical features of high fever, skin rash, headache, myalgia, lymphadenopathy, gastrointestinal symptoms, and cough. Most patients have a relatively benign course. A typical eschar occurs at the site of the bite but is not always present. According to organ involvement, patients could present with pneumonitis, meningitis, encephalitis, myocarditis, acute pulmonary edema, pericarditis, hepatitis, or multiorgan failure.⁴⁶¹ Severe disease is characterized by acute respiratory distress syndrome, AKI, bleeding, coagulation disorders, meningoencephalitis, and shock.⁴⁶²⁻⁴⁶⁶ Respiratory distress and encephalitis are the major causes of death.⁴⁶⁷ Renal involvement is common. Urinary abnormalities, such as hematuria, proteinuria, pyuria, and granular casts, occur in up to 80% of patients, and AKI occurs in 8% to 60%.^{463-466,468-472} Pathogenic mechanisms include direct renal tubular cell injury with or without tubulointerstitial inflammatory cell infiltration.^{473,474} A higher serum level of neutrophil gelatinase-associated lipocalin has been shown to be associated with the development of AKI in scrub typhus.⁴⁷⁵ Intravascular hemolysis in patients with glucose-6-phosphate dehydrogenase deficiency, rhabdomyolysis, hypoperfusion, and vasculitis are other contributing factors to kidney injury.^{468,476} Histopathology shows acute tubular necrosis, interstitial nephritis, and mesangial proliferation.^{473,477-479} A retrospective study of 510 Korean patients with scrub typhus diagnosed in 2001–2013 showed that 35.9% developed AKI and that risk factors for AKI included older age and concomitant comorbidities such as diabetes mellitus and preexisting CKDs.⁴⁸⁰ Furthermore, the complication of AKI was associated with higher rates of intensive care unit admission and death. All except one patient with AKI recovered. Diagnosis of acute scrub typhus infection can be reached by PCR to detect bacterial DNA, immunofluorescence assay for IgM antibody, or Weil-Felix test, which has a lower sensitivity but is still useful in resource-limited areas.⁴⁸¹⁻⁴⁸³ Prompt treatment with doxycycline results in rapid improvement including that of renal function.^{464,477} Elderly patients are at risk of more

complications and poor clinical outcome, which could also be related to delayed presentation.⁴⁸⁴

SEVERE FEVER WITH THROMBOCYTOPENIA SYNDROME

Severe fever with thrombocytopenia syndrome (SFTS) was initially recognized in the Henan and Hubei provinces of China around 2007⁴⁸⁵ and was subsequently reported in other provinces in China such as Zhejiang and Liaoning,^{486,487} South Korea,⁴⁸⁸ and Japan.⁴⁸⁹ The syndrome is caused by the SFTS phlebovirus, officially named Dabie bandavirus, which belongs to the genus bandavirus in the family Phenuiviridae, order Bunyvirales, and ticks (e.g., *Rhipicephalus microplus*) in domestic animals are the vector.^{485,490-494} Investigations showed that SFTSV was common throughout different parts of Korea and China, especially in the warmer areas where ticks were more prevalent, and the disease mostly occurred in the months between April and October with farmers at higher risk due to occupational exposure.^{492,495-497} A report from South Korea suggested that deforestation contributed to outbreaks of the infection.⁴⁹⁸ A retrospective study in Jiangsu, China, showed an increasing incidence rate over the period 2011–2022, with a total of 698 cases and 6% mortality rate.⁴⁹⁹ Patients infected with SFTSV presented with fever, thrombocytopenia, leukocytopenia, hemorrhagic features, and in severe cases multiorgan failure. Mortality rate was reported as 8.7% to 47.2%.^{496,500} AKI occurred in about 20% of cases and was associated with a higher mortality rate of 12% to 30%.⁴⁸⁵ Studies from China reported hemorrhagic features in >70% of patients including gastrointestinal bleeding in 10% of patients, liver dysfunction in the majority, AKI requiring dialysis in 23%, and a 27% mortality rate.^{501,502} In a study that included 129 cases from Zhejiang province, China, the death rate was 12.4% and extremes of body mass index and delayed diagnosis were associated with increased risk of death.⁵⁰³ Clinical features of SFTS are nonspecific and overlap with those of human granulocytic anaplasmosis (caused by *Anaplasma phagocytophilum*), HFRS due to leptospirosis, or hantavirus infection,⁴⁹⁵ while lymphadenopathy has been reported as more common in SFTSV and AKI more common in hantavirus infection (>80%).⁵⁰⁴ CD3- and CD4-positive T lymphocytes were lower than normal in patients, while natural killer cell counts were increased.⁵⁰⁵ Impaired β -interferon production and induction of a proinflammatory cytokine storm contributed to disease pathogenesis.^{486,506} Nonstructural proteins of SFTSV could suppress exogenous type I interferon-induced Janus kinase/signal transducer and activator of transcription (Jak/STAT) signaling to facilitate the virus evading host immune surveillance.⁵⁰⁷ Diagnosis can be made by serologic testing with ELISA or detection of viral genome using reverse transcription–PCR or reverse transcription–cross-priming amplification.⁵⁰⁸⁻⁵¹¹ Supportive therapy and treatment of various complications are currently the mainstay of management.⁴⁸⁷ Treatment with ribavirin has been reported to reduce fatality rate from 6.25% to 1.16% in patients with a low viral load $<1 \times 10^6$ copies/mL.⁵¹²

HANTAVIRUSES AND HEMORRHAGIC FEVER WITH RENAL SYNDROME (HFRS)

Hantaviruses are single-stranded, enveloped, negative-sense RNA viruses that belong to the Bunyaviridae family.⁵¹³

Hantavirus is named after the Hantaan River in South Korea around which an outbreak occurred. The virus was isolated in the late 1970s.⁵¹⁴ The zoonosis is endemic in East Asia including China, Korea, and part of Russia.⁵¹⁵⁻⁵¹⁷ Humans may become infected with hantaviruses through aerosolized excreta inhalation and contact infection by rodent urine, saliva, or feces, and human-to-human transmission has been reported with the Andes virus in South America.⁵¹⁸ Some strains of hantaviruses cause potentially fatal diseases in humans, such as hantavirus HFRS and hantavirus pulmonary syndrome—also known as hantavirus cardiopulmonary syndrome.⁵¹⁴

HFRS (also termed Korean hemorrhagic fever, epidemic hemorrhagic fever, or nephropathia epidemica) is a group of clinically similar illnesses caused by species of hantaviruses. HFRS can thus be caused by the Hantaan virus (HTNV), Amur virus (AMV), Seoul virus (SEOV), Dobrava virus (DOBV), or Puumala virus (PUUV), with variable disease severity (often more severe with HTNV and DOBV).⁵¹⁹ HTNV and SEOV are found in China, with the main natural hosts being *A. agrarius* and *Rattus norvegicus*, respectively.⁵²⁰ SEOV is globally widespread and may cause moderate HFRS, whereas PUUV is endemic in northern Europe and may cause a generally mild form of HFRS, also called “nephropathia epidemica.” About 100,000 cases of HFRS are documented annually,⁵²¹ most of which are in China, Korea, and Russia, though the disease has also been reported in Europe, Africa, and the Americas. PUUV is found in Scandinavia, Western Europe, and western part of Russia. DOBV is found primarily in the Balkans, and SEOV is found worldwide. Saaremaa virus is found in central Europe and Scandinavia.

Hantavirus infects human vascular endothelial cells and damages capillaries and small vessels. Viral entry is mediated via integrins and CD55.⁵²² Immune-mediated mechanisms including cytokines and complement activation are involved in organ injury.⁵²³⁻⁵²⁵ Clinical manifestations include AKI, increased vascular permeability, and coagulation abnormalities. Typical kidney biopsy findings include acute tubulointerstitial nephritis, hemorrhage into medullary tissues, interstitial edema, renal tubular cell necrosis, and in some cases, podocyte injury with foot process effacement associated with mislocalization of junctional proteins, thereby resulting in glomerular proteinuria.^{526,527} Hantavirus-infected vascular endothelial cells were shown to be sensitized to the action of vascular endothelial growth factor in increasing permeability.⁵²⁸ AKI was associated with increased mortality, especially in elderly patients who also had more severe hypotension.⁵²⁹ In HFRS, the clinical picture is characterized by AKI, often with massive proteinuria due to tubular and glomerular involvement.⁵¹³ Infection with hantavirus is associated with significant morbidity and mortality worldwide. Approximately a quarter of infected patients have a severe course. A mortality rate of 5% to 10% has been reported from Korea, whereas in Europe the mortality rate appeared lower at 0.5%.⁵¹³

Symptoms usually develop 1 to 2 weeks after exposure to infectious material but can be delayed for weeks. Initial symptoms begin suddenly and include severe headache, back, and abdominal pain, fever, chills, nausea, facial flushing, conjunctival injection, and skin rash. Severe manifestations include hypotension, vascular leakage, and AKI.

HFRS follows a typical clinical course (Table 78.3).^{513,530} In the initial “febrile phase,” which lasts for 3 to 7 days, symptoms often include facial redness, fever and sweating, headache, malaise, abdominal and back pain, nonspecific respiratory symptoms, and gastrointestinal symptoms such as nausea and diarrhea. This is followed by the “hypotensive and hemorrhagic phase,” lasting for about 2 days, which is associated with increased vascular permeability. Tachycardia occurs and patients can develop hypoxemia. Leukocytosis occurs with thrombocytopenia, elevated C-reactive protein, and lactate dehydrogenase levels. The next stage is the “kidney phase,” which lasts 3 to 7 days, characterized by AKI with frequent oliguria, hematuria, proteinuria, and hypoalbuminemia. The severity of AKI has been associated with the severity of thrombocytopenia and also the presence of glomerular proteinuria.^{513,529,531} This initial oliguria is followed by subsequent polyuria and convalescence.

Diagnosis is by serologic testing of IgM against hantavirus by various means such as reverse transcription–PCR.^{532,533} Supportive therapy is still the mainstay of care for patients with hantavirus infection.⁵³⁴ Careful attention to fluid management and hemodynamics is necessary, with RRT by intermittent HD or continuous RRT in unstable subjects. The role of ribavirin remains controversial, with results showing a lack of efficacy and safety.⁵³⁵⁻⁵³⁷

DIALYSIS IN ASIA

Dialysis, while lifesaving for patients with ESKD, is a high-cost treatment, and the access to dialysis in developing countries or those with less developed national health care insurance systems is fraught with economic and ethical issues. Over the period of 2000–2013 the incidence of ESKD increased 13 times in Thailand, 2 to 3 times in the Philippines, Malaysia, and South Korea, and the reported figures were affected by the variable improvements in the access to RRT between different countries.⁵³⁸ While more than 1 million people are on maintenance dialysis in Asia,

the prevalence varies considerably within the region, owing to differences in the status of development of health care systems in different regions. Access to dialysis therapy is not restricted by personal financial means in Japan, Taiwan, South Korea, Singapore, and Hong Kong, and there is ongoing expansion of health care insurance coverage for dialysis in other places such as Thailand, Malaysia, and many parts in China. Dialysis registries have been established in Japan, South Korea, Taiwan, Malaysia, Hong Kong, Shanghai, Thailand, and Singapore.⁴⁶ The incidence and prevalence rates of treated ESKD are highest in Japan, Taiwan, South Korea, and Singapore, at approximately 300 to 450 pmp and 1800 to 3000 pmp, respectively, and are much lower in areas with lower income such as the Philippines and Indonesia.⁵³⁸ With the size of the ESKD population predicted to increase to 5.4 million by 2030, the financial implications of the RRT program are daunting.²⁵

Health care policies have evolved in response to local socioeconomic circumstances. In appropriately chosen patient populations, PD and HD are associated with comparable short- and medium-term survival and quality of life outcomes, while the relative costs of the two dialysis modalities vary considerably between geographic regions due to multiple factors such as differing costs of labor and transportation. Comparative data on long-term survival between PD and HD are confounded by selection bias and high drop-out rates. Calculation of the overall health care cost also needs to consider the costs associated with technique failure and change to an alternative dialysis modality, as well as the costs associated with management of related complications such as encapsulating peritoneal sclerosis in patients on long-term PD. A cost-effectiveness study from Japan published in 2015 showed that, based on 29 maintenance HD patients (8 had diabetes mellitus) with a mean age of 59.9 years, the estimated total medical cost for HD treatment was approximately US\$45,200 annually, and the incremental cost-utility ratio was $6.88 \pm 4.47 \times \text{US\$104}$ per quality-adjusted life-year (QALY), being higher for patients with diabetes mellitus ($8.17 \pm 6.28 \times \text{US\$104}$) per QALY.⁵³⁹ A retrospective longitudinal study

Table 78.3 Typical Clinical Phases of Hemorrhagic Fever with Renal Syndrome due to Hantavirus Infection

Phase of Illness	Time of Occurrence	Principal Features	Signs and Symptoms	Complications
1. Febrile	1-7 days	Fever	- Headache - Vomiting - Abdominal pain - Back pain - Visual disturbance	- Acute encephalomyelitis - Bleeding - Shock - Multiorgan failure - Pituitary hemorrhage
2. Hypotensive	1-3 days	Hypotension	- Capillary leak - Pulmonary symptoms	- Renal - Pulmonary edema
3. Oliguric	2-6 days	↓ Urine output	- Severe oliguria can occur - Fluid retention	Acute respiratory distress syndrome
4. Polyuric	2 weeks	↑ Urine output	Decrease of body weight	Disseminated
5. Convalescent	3-6 months		- Weakness - Fatigue	intravascular coagulation

Modified from Jiang H, Du H, Wang LM, et al. Hemorrhagic fever with renal syndrome: pathogenesis and clinical picture. *Front Cell Infect Microbiol.* 2016;6:1.

in Taiwan showed that health care policy initiatives were effective in increasing the PD utilization rate, although this was accompanied by increased rates of technique failure and mortality.⁵⁴⁰ A report from Hong Kong, which has adopted a PD-first policy since the mid-1980s, showed that the overall annual health care and societal cost locally in the first and second years after initiation of dialysis in 2017 was highest for hospital-based HD ($\approx$ US\$55,460/year) and lowest for PD ($\approx$ US\$21,800/year), with nocturnal home HD in between ($\approx$ US\$22,300/year).⁵⁴¹ Using Markov modeling to investigate lifetime cost-effectiveness under the PD-first policy, the same team of investigators showed more QALYs of 8.37, 7.13, and 6.58 life-years for home-based HD, PD, and hospital-based HD, respectively, while the cost for hospital-based HD and home-based HD was 1.85- and 1.27-fold that of PD, respectively.⁵⁴² A PD-first policy with universal health coverage was introduced in Thailand in 2008. A cost-utility analysis based on updated real-practice data showed that PD-first policy remained more cost-effective compared with HD-first policy, but HD gained 1.05 to 1.76 more QALYs compared with PD.⁵⁴³ Starting from February 2022 the universal coverage in Thailand has extended to include HD. While cost is often a focus of attention from the perspective of health care administrators, other factors come into play when patients select a dialysis modality. A Korean multicenter study reported that of 2514 incident dialysis patients, 69.5% selected HD and 30.6% selected PD, respectively, and that PD was more commonly selected by patients younger than 65 years old (37.1 vs. 16.9%), with planned dialysis being an important factor contributing to its selection.⁵⁴⁴ One survey reported that the incidence rate (per million population) of kidney replacement therapies in this region was highest in Taiwan at 493, while Brunei, Japan, Singapore, South Korea and Thailand had rates of >300 . The prevalence rate (per million population) of kidney replacement therapies was 3392 in Taiwan, 2599 in Japan, 1816 in South Korea, and >1300 in Brunei, Hong Kong, Malaysia, Singapore, and Thailand.⁵⁴⁵

PERITONEAL DIALYSIS

Insufficient HD facilities, draconian selection criteria for RRT, cost containment, and clinical outcomes that were at least comparable with HD in the short term were factors that led to the establishment of a PD-first policy for publicly funded dialysis service in Hong Kong in the mid-1980s, and to this date, Hong Kong continues to report the highest PD utilization rate in the world ($\approx 75\%$ of all dialysis patients).⁵⁴⁶⁻⁵⁴⁹ Thailand also had a PD-first policy from 2008 to 2022. A study from Indonesia based on economic modeling reported an equal number of 5.93 life-years saved by either a PD-first or an HD-first policy, while a PD-first policy would be associated with a slightly higher QALYs of 4.40 (compared with 4.34 for HD-first), a lower total lifetime treatment cost of 700 million IDR (US\$51,800) per patient (compared with US\$54,400 for HD-first), and a substantial health care budget saving in the long term.⁵⁵⁰ “Cost-effectiveness,” merits and perceptions by patients, and feasibility of dialysis modality policies, however, vary between health care settings. PD would be more suitable than HD in patients with unstable cardiovascular status or vascular access problems or those who live far from dialysis

facilities. Care should be exercised to ensure adequate dialysis dose and ultrafiltration in patients of large body size, especially those with minimal residual kidney function.⁵⁵¹ PD outcomes have improved over the decades but with considerable variability.⁵⁵² Issues related to prolonged PD are pertinent concerns in Asian countries with a low rate of kidney transplantation. Despite the advent of PD solutions with improved biocompatibility,^{553,554} progressive peritoneal membrane fibrosis and deterioration of dialytic functions of the peritoneum remain unresolved problems limiting the long-term application of PD and also having an adverse impact on patient outcomes.⁵⁵⁵⁻⁵⁵⁷ Encapsulating peritoneal sclerosis is an uncommon but severe complication associated with a high mortality rate and can present even after kidney transplantation.⁵⁵⁸ Duration of PD is a significant risk factor because encapsulating peritoneal sclerosis predominantly affects patients on PD for >5 years, and the incidence rate increases with further time on PD.⁵⁵⁹ Evidence suggests that the use of neutral pH low-GDP PD solutions, in contrast to the early PD fluids with acidic pH and high GDP, is associated with reduced incidence of encapsulating peritoneal sclerosis, likely due to reduced peritoneal injury inflicted by the PD solution, while peritonitis remains a major contributing factor.⁵⁶⁰

HEMODIALYSIS

HD service in Japan is characterized by a high proportion of in-center HD ($>85\%$), which is related to the progressive establishment of accessible HD centers over recent decades and clinical outcomes that are deemed favorable by international standards including both early and long-term survival rates.⁵³⁸ The low rate of kidney transplantation in Japan has contributed to the progressive expansion of the dialysis population, especially the elderly age group and those with diabetes mellitus.⁵⁶¹ Unadjusted 5-year survival rate was 60% in Japan compared with 39% in the United States in 2004–2008.⁵⁶² Studies from the DOPPS were launched in the United States in 1996 and extended to Japan in 1998–2002 and China in 2010. Results from DOPPS and Japan Dialysis Outcomes and Practice Patterns Study (J-DOPPS) showed that, compared with Western countries, HD patients in Japan showed a high utilization rate ($>90\%$) of arteriovenous fistula, less intensive anemia management with lower-target hemoglobin, and lower doses of erythropoietin-stimulating agent, as well as IV iron, lower parathyroid hormone levels, and lower levels of C-reactive protein. Treatment time was >4 hours in approximately 80% of Japanese patients, compared with 40% in the United States.^{538,563} Similar to earlier findings in the Hemodialysis Study and DOPPS, results from the MONDO database, which included data of HD patients from different networks globally including the Asian-Pacific region, showed that failure to convert from central venous catheter access to noncatheter vascular access within 6 months of commencing HD was associated with a higher mortality rate.⁵⁶⁴⁻⁵⁶⁶ Twice-weekly HD is not uncommon in Asian countries, especially those with insufficient access to HD or where patients are required to pay for the treatment. In this regard, use of twice-weekly HD is $<3\%$ in Japan, Europe, and North America but is approximately 20% in China.⁵⁶⁷ Data from serial phases

of J-DOPPS also showed a U-shaped relationship between blood pressure and all-cause mortality, with lowest mortality in patients with baseline systolic blood pressure of 140 to 159 mm Hg and diastolic blood pressure of 65 to 74 mm Hg,⁵⁶⁸ an association between vascular access failure and high intradialytic ultrafiltration (HR: 1.41 when ultrafiltration was 5.1%–13.7% of body weight),⁵⁶⁹ an inverse relationship between employment and mortality or hospitalization,⁵⁷⁰ and improved prediction of cardiovascular events using a new model, which incorporated six variables (age, diabetes mellitus, history of cardiovascular events, dialysis time per session, serum phosphorus level, and serum albumin level) compared with the Framingham Risk Score.⁵⁷¹ Data on 1350 patients from the China Dialysis Outcomes and Practice Patterns Study (C-DOPPS) showed that 21% had hemoglobin <9 g/dL and attributed the relatively high percentage to less frequent HD and financial barriers.⁵⁷² Recent J-DOPPS data showed that surrogate indicators of nutritional status such as the Geriatric Nutritional Risk Index and Creatinine Index had an inverse relationship with mortality risk of patients on long-term hemodialysis.⁵⁷³

KIDNEY TRANSPLANTATION—SPECIAL ISSUES IN ASIA

KIDNEY TRANSPLANTATION ACROSS ABO BLOOD GROUP BARRIERS

Due to the rarity of deceased-donor organ transplantation, kidney transplantation across ABO blood group barrier has developed in Japan since the late 1980s. The first case of ABO-incompatible (ABO-I) kidney transplantation from a live donor in Japan was reported in 1991.⁵⁷⁴ Subsequently, >3500 ABO-I kidney transplantations have been performed, and the clinical outcome is now comparable with ABO-compatible (ABO-C) procedures.^{575–577} In particular, continued improvements in preoperative desensitization therapy, which prevents posttransplant serious acute antibody-mediated rejection, have contributed to its successful development.^{577,578} ABO-I kidney transplantation programs are now well established in many transplant centers in Japan and account for approximately 30% of live-donor kidney transplant operations in Japan.⁵⁷⁷ The numbers of centers and patients undergoing ABO-I kidney transplantation have also increased rapidly in Korea, and ABO-I kidney transplantation accounted for 10% and >20% of all live-donor kidney transplantations in 2010 and 2014, respectively.^{579–581} ABO-I kidney transplantation is also performed in selected patients in Thailand, Malaysia, China, and Hong Kong.

Anti-blood group antibody-mediated acute rejection ceases to appear after accommodation is induced a few weeks after kidney transplantation.⁵⁸² Therefore the most important elements in the pretransplant desensitization protocol are removal of preformed antibodies and inhibition of antibody production by B cells. Previously, this could only be achieved with splenectomy.^{583,584} Rituximab has been used in ABO-I kidney transplantation in Japan since 2002 and is now an integral element in standard desensitization protocols.^{585,586} The incorporation of tacrolimus and mycophenolate into

the immunosuppressive regimen has also contributed to improved clinical outcomes.⁵⁸⁷ The 5-year graft survival rates of ABO-I kidney transplantation in Japan improved from 68% in 1989–1994 to 90% in 2001–2006 and further improved after the introduction of rituximab.^{576,578} The 5-year graft survival rates of ABO-C, ABO-I with splenectomy, and ABO-I with rituximab were reported as 88.4%, 90.3%, and 100%, respectively.⁵⁸⁸ An analysis of single-center data from 1032 live-donor kidney transplantations performed in 1989–2013, of whom 247 were ABO-I, showed that ABO-I and ABO-C cases had comparable 9-year graft survival rates (86.9% and 92.0%, respectively) and that the graft survival was better in 2005–2013 compared with 1998–2004 (HR: 0.30).⁶⁴

A desensitization protocol comprising mycophenolate mofetil and low-dose corticosteroids started 28 days pretransplant, preoperative plasmapheresis, 2 doses of rituximab at 375 mg/m² on day –14 and day –1, and postoperative immunosuppression with basiliximab, corticosteroids, and tacrolimus or cyclosporine was investigated in a prospective 1-year study. It included 18 patients who underwent ABO-I kidney transplantation, and the results showed high tolerability of the treatment protocol and high efficacy with no anti-A/B antibody-mediated rejection.⁵⁸⁹ One patient developed anti-HLA antibody-mediated rejection, which was successfully managed with standard antirejection therapy. Patient and graft survival were both 100% after 1 year. In a retrospective series of 327 Japanese patients who underwent live-donor kidney transplantation, comparison of protocol graft biopsies performed at 3 and 12 months after transplantation between 226 ABO-C and 101 ABO-I patients showed comparable allograft histology overall, with subclinical acute rejection detected in 6.9% and 9.9% of patients in the ABO-C and ABO-I groups at 3 months ($P = .4$) and in 12.4% and 10.1% at 12 months, respectively ($P = .5$). No between-group difference with regard to the rate of infections (including BK nephropathy) and patient or graft survival was found.⁵⁹⁰ Latest developments include investigations into whether low-dose rituximab 100 mg for once may be sufficient for patients with relatively low titer of anti-A/B antibodies of below 1:32⁵⁹¹ and the mechanisms of immunologic hyporesponsiveness to blood group antigens expressed in kidney cells.⁵⁹²

Worldwide, the Collaborative Transplant Study reported outcomes from 1420 ABO-I kidney transplantations performed at 101 transplant centers from 2005 to 2012.⁵⁹³ The 3-year graft survival rate of ABO-I transplants did not differ from matched ABO-C cases (89.9% vs. 90.1%). Patient survival in ABO-I cases was lower initially due to infection-related deaths, but 3-year patient survival rates were similar between the two groups (95.6% in ABO-I vs. 96.3% in ABO-C). Data from 125 Korean patients who underwent ABO-I kidney transplantation from 2007 to 2010 showed 2-year graft and patient survival rates of 97.5% and 99.2%, respectively.⁵⁷⁹

HEPATITIS B IN KIDNEY TRANSPLANT RECIPIENTS

One-third of the world's population shows serologic evidence of HBV infection, and approximately 300 million people are chronic HBV carriers, with a worldwide seroprevalence

rate of 3.61% for HBsAg.⁵⁹⁴⁻⁵⁹⁶ Endemicity of HBV infection varies among Asian countries, with HBsAg seroprevalence rates ranging from <1% to >10% (Table 78.4).⁵⁹⁶

Hepatitis B vaccination and matching of HBV status between donor and recipient are measures to prevent HBV infection in kidney transplant recipients. Hepatitis B vaccination at birth was introduced in some Asian countries in the late 1980s but is still not universally practiced, especially in less developed areas. EASL clinical practice guidelines recommend vaccination in HBV-seronegative patients with end-stage renal disease.⁵⁹⁷ The efficacy of hepatitis B vaccination is significantly lower in patients with moderate to severe renal impairment and in immunosuppressed patients,^{598,599} and thus patients with CKD should receive vaccination early. Maintaining an anti-HBs titer of ≥ 100 IU/L with booster is in general recommended, although hepatitis B reactivation could still occur in anti-HBc-positive subjects upon heavy immunosuppression.^{600,601} Donor-recipient matching with regard to HBV serologic status aims to prevent transmission of HBV from the kidney donor to the recipient. Kidney from an HBsAg-positive donor is normally not transplanted into a recipient who is seronegative for HBsAg, anti-HBs, and anti-HBc, except when the transplant is deemed urgent and antiviral treatment is available. The risk of HBV transmission from HBsAg-negative anti-HBc-positive donors to HBsAg-negative recipients appeared low, especially when the recipient is anti-HBs-positive,⁶⁰² and transplanting a kidney from an HBsAg-positive donor into an anti-HBs-positive recipient with immunoglobulin cover appeared safe,⁶⁸ although in both cases surveillance of de novo HBV infection with HBsAg and HBV DNA monitoring is advisable.

Immunosuppression enhances viral replication, leading to reactivation of hepatitis B and liver disease that can be life-threatening.^{603,604} Before the availability of antiviral prophylaxis, the rates of HBV reactivation in HBsAg-positive kidney transplant recipients were reported as 50% to 94%, and HBV infection was associated with significantly increased morbidity and mortality, related to liver complications and nonliver complications, especially infections.^{66,603,605-608} Clinical manifestations of HBV-related liver complications after kidney transplantation can present as acute hepatitis,

fibrosing cholestatic hepatitis, chronic hepatitis with fulminant reactivation, chronic hepatitis, cirrhosis and its complications, or hepatocellular carcinoma.^{66,67,608-613} The 10- and 20-year patient survival rates in HBsAg-positive kidney transplant recipients without antiviral therapy were 85% and 71%, respectively (vs. 98% and 95% at 10 and 20 years in HBsAg-negative patients).⁶⁷ Before the availability of antiviral treatment, HBsAg-positive kidney transplant recipients showed a 9.7-fold increased risk of death and 68-fold increase in liver-related mortality rate compared with HBsAg-negative controls, while the survival of these patients is similar to subjects without HBV infection in the present era of effective antiviral nucleoside/tide analog therapies.⁶⁶

Interferon as a therapy for hepatitis B precipitates allograft dysfunction and rejection and is thus contraindicated in kidney allograft recipients except for lifesaving indications.⁶¹⁴ The advent of effective antiviral nucleoside/tide analog therapies has markedly improved posttransplant outcomes of patients infected with HBV. Antiviral treatment can be given prophylactically at the time of kidney transplantation or preemptively when increasing levels of circulating HBV DNA indicate impending reactivation, although the former is more convenient, especially in the context of a busy clinical setting. Salvage therapy, given after the onset of clinically evident liver exacerbation, is associated with inferior outcomes.⁶¹⁵ A study that included 26 HBsAg-positive Chinese patients who received lamivudine therapy preemptively based on increasing HBV DNA levels showed effective suppression of viral replication and normalization of liver biochemistry and improved survival rate to that approaching HBsAg-negative patients.⁶⁶ The high incidence of drug resistance associated with lamivudine was tackled with newer antiviral agents including adefovir, entecavir, and tenofovir.⁶¹⁶⁻⁶¹⁸ Adefovir is nephrotoxic, and its role has been superseded by newer agents. At present, entecavir and tenofovir are the recommended first-line antiviral therapies.²⁹⁹ Entecavir is preferred in the kidney transplant setting, especially in patients with suboptimal graft function as tenofovir is associated with renal toxicity.⁶¹⁹ A 20-year patient survival rate of 69% has been reported for HBsAg-positive kidney transplant recipients transplanted in the recent era of antiviral treatment, which was superior to 40% in patients

Table 78.4 HBsAg Seroprevalence and the Number of People Living with Chronic Hepatitis B Virus in the General Population in Some Asian Countries

Country	Prevalence Estimates (% , 95% CI)	Population Size Per Country	HBsAg-Positive Population
China	5.49 (5.47-5.50)	1,359,821,465	74,601,204
Japan	1.02 (1.01-1.02)	127,352,833	1,294,431
Malaysia	0.74 (0.70-0.77)	28,275,835	208,540
Mongolia	9.07 (8.41-9.78)	2,712,738	246,070
The Philippines	4.63 (4.53-4.73)	93,444,322	4,326,212
Singapore	4.09 (3.87-4.33)	5,078,969	207,943
South Korea	4.36 (4.36-4.37)	48,453,931	2,111,914
Thailand	6.42 (6.37-6.47)	66,402,316	4,260,008
Vietnam	10.79 (10.29-11.31)	89,047,397	9,607,438

CI, Confidence interval; HBsAg, hepatitis B surface antigen; HBV, hepatitis B virus.

Modified from Schweitzer A, Horn J, Mikolajczyk RT, et al. Estimations of worldwide prevalence of chronic hepatitis B virus infection: a systematic review of data published between 1965 and 2013. *Lancet*. 2015;386:1546–1555.

transplanted before the advent of antiviral drugs.⁶¹⁷ Overall, the short- and medium-term survival rates of HBsAg-positive kidney transplant patients are now comparable with subjects not infected with HBV.^{66,620,621} Nevertheless, while antiviral therapy effectively prevents severe exacerbation due to immunosuppression, liver-related deaths can still occur during long-term follow-up due to progressive liver disease. Data from a retrospective study of 74 Korean patients who had received antiviral treatment showed comparable renal allograft survival between HBsAg-positive and HBsAg-negative patients, while seropositivity for HBsAg was associated with 2.19-fold increased risk of death, with one-third of the deaths related to HBV, during a mean follow-up of 75.7 months.⁶²² In another multicenter study that included 160 Korean kidney transplant recipients with HBV infection, transplanted after 1999, HBV infection was associated with 2.37-fold increased risk of death and mortality correlated with the severity of inflammation in the liver biopsy before kidney transplantation.⁶²³ Surveillance measures including monitoring of liver function and the development of malignancy by liver ultrasound and alpha-fetoprotein measurement are thus important elements in the clinical care of kidney transplant recipients infected with HBV in Asia.

B-cell depletion with rituximab is often included in desensitizing immunosuppressive protocols for patients with anti-blood group or anti-HLA antibodies before kidney transplantation.⁶²⁴ However, rituximab is associated with a high rate of hepatitis B reactivation in not only HBsAg-positive subjects but also subjects with apparent clinical resolution of prior infection as denoted conventionally by seronegativity for HBsAg and seropositivity for anti-HBc.⁶²⁵⁻⁶²⁸ The rate of hepatitis B reactivation in HBsAg-negative/anti-HBc-positive patients who received rituximab-containing chemotherapy was reported as 11.3% to 41.5%.^{629,630,631} Patients with autoimmune diseases treated with rituximab are similarly at risk.^{632,633} In this regard, a study in 49 HBsAg-negative anti-HBc-positive kidney transplant recipients in Korea showed hepatitis B reactivation in 5 patients (10.2%), in whom 2 it was regarded as clinically severe, resulting in 1 mortality.⁶³⁴ Hepatitis B reactivation may also occur after costimulation blockade with belatacept, which has been used for calcineurin inhibitor sparing after kidney transplantation, and a reactivation rate of 21.9% in anti-HBc-positive kidney transplant recipient has been reported including in patients who were originally HBsAg-negative.^{635,636}

Before the advent of effective and safe antiviral treatment, patients with cirrhosis were largely declined for kidney transplantation in view of the unacceptably high rate of liver-related mortality.⁶³⁷ In the present era, patients with compensated cirrhosis and no evidence of portal hypertension can be considered for kidney transplantation. Data from a retrospective study in Korea of 12 kidney transplant recipients with prior HBV-associated liver cirrhosis showed liver deterioration after kidney transplantation in one patient and stable liver status in the other 11 patients.⁶³⁸ Patient survival rate at 5 years was similar in patients with or without prior cirrhosis, at 100% and 94%, respectively. Patients with decompensated cirrhosis and those with compensated cirrhosis and portal hypertension are not eligible for kidney transplantation

but may be considered for combined liver and kidney transplantation.⁶³⁷

PEDIATRIC NEPHROLOGY

Recent reports have shed light on the development and status of pediatric nephrology in the region. A retrospective study in Taiwan that included 339 pediatric kidney biopsies in the period 2002–2020 showed that the mean age of diagnosis for children with glomerular diseases was 13.7 years with a female-to-male ratio of 1.19:1, and 55.7% of patients presented with nephritic syndrome, while 22.7% presented with nephrotic syndrome and 13.9% with asymptomatic hematuria.⁶³⁹ Proliferative lupus nephritis (41%), minimal change disease (18%), and IgAN (10%) were leading diagnoses, and poststreptococcal glomerulonephritis was infrequent and never diagnosed after 2010. On the basis of information from the Japan Renal Biopsy Registry within the period 2007–2017 on 3463 patients below the age of 15 years and 6532 patients aged 15 to 30 years, investigators found that nephrotic syndrome was the leading clinical presentation in children younger than the age of 4 years, and IgAN was the most common histologic diagnosis in young adults.⁶⁴⁰ A nationwide survey of 563 Japanese children aged 1 to 15 years with Henoch-Schonlein purpura nephritis showed an annual incidence rate of 1.32/100,000, with mean presentation at age 7 and a male-to-female ratio of 1.2:1.⁶⁴¹

Regarding the more common glomerular diseases in children, it is well recognized that children with SLE often have more severe disease compared with adults. Despite the severe disease, one report showed a complete/partial remission rate of 78% and 8%, respectively, after 12 months of treatment and a 3.2% ESKD rate after median follow-up of 10.3 years, with 2 deaths among a total of 92 patients, thus highlighting the efficacy of therapy in the present era and the importance of early treatment.⁶⁴² Favorable long-term results have been reported in childhood-onset lupus nephritis treated with triple immunosuppression that included corticosteroids, mycophenolate, and cyclosporine.⁶⁴³ Similarly, timely and individualized use of immunosuppression was associated with favorable outcomes in pediatric IgA vasculitis nephritis.⁶⁴⁴ A retrospective analysis of 90 Chinese children with IgAN and nephrotic-range proteinuria showed that 21% had impaired kidney function at presentation, among whom 73.7% recovered to normal level after treatment.⁶⁴⁵ All except two patients were treated with corticosteroids with or without immunosuppressants, and complete or partial remission of proteinuria was achieved in 88.6% and 10.1%, respectively. CKD occurred in 7.8% of patients on follow-up. Infection is an important complication in children with idiopathic nephrotic syndrome. A retrospective study of 187 pediatric patients with idiopathic nephrotic syndrome (85 steroid-dependent frequent relapsers; 45 steroid-resistant) in Japan showed that infections occurred in 55 patients (29.4%).⁶⁴⁶ Unlike previous studies reporting that peritonitis and bacteremia were major causes of infection in this population, pneumonia was most common (21 of 84 infection episodes) and rituximab exposure was identified as a significant risk factor. In pediatric patients with relapsing

idiopathic childhood nephrotic syndrome, there is evidence to suggest that repeated courses of rituximab can reduce the risk of relapse and is in general well tolerated.⁶⁴⁷ In 346 children who had received 1149 courses of rituximab in total, hypogammaglobulinemia occurred in 50.9% with one-fifth considered clinically significant and infection occurred in 4.5%.

There are relatively little data on AKI in children. In a retrospective analysis of 1678 children admitted to six pediatric intensive care units in China, it was found that the Pediatric Reference Change Value Optimized for Acute Kidney Injury in Children (pROCK) definition of AKI, which took into account the variation of serum creatinine with age, was associated with better separation of AKI stages and correlation with outcome compared with pRIFLE and KDIGO classification.⁶⁴⁸ A report from Hong Kong showed that AKI was a common complication in children with hematologic malignancies, affecting 26.9% of 130 patients (32 episodes per 100 patient-years), and 85.3% occurred during chemotherapy (29.2% related to septic shock). Also, the occurrence of AKI was associated with inferior 12-month patient survival (77.1% vs. 94.7%, $P = 0.002$).⁶⁴⁹

Childhood- or adolescent-onset CKD has a major impact on patients and families. A territory-wide study spanning 20 years in Hong Kong reported an incidence rate for ESKD of 6.28 per million age-related pediatric population, with 33% due to congenital abnormalities, and the median time to receive a kidney transplant was 3.7 years while the 10- and 15-year survival rate was 89.7% and 87.1%, respectively (72% due to infections).⁶⁵⁰ While survival and clinical outcomes of pediatric patients with ESKD have improved, a recent report from Japan showed a high prevalence of depression in 29% of patients aged 16 to 39 years who developed ESKD before the age of 20 years, while their SF-36 physical and mental components were comparable with controls.⁶⁵¹ In this context, the PedsQL 3.0 End Stage Renal Disease Module is an instrument used internationally to assess health-related quality of life (HRQoL) in children with CKD, and a Malay language version (Bahasa Melayu) was recently validated.⁶⁵² Also, a report from Japan highlighted the importance of end-of-life care in children who are excluded from kidney transplantation due to severe extrarenal complications such as psychomotor retardation or cardiac complications.⁶⁵³

CONCLUSION

Clinical nephrology has advanced significantly over the past half century in the Far East. Variations in geotopography and genetic background, as well as rapid changes in the urban environment and socioeconomic development, have contributed to the marked diversity of disease spectra and multiplicity of challenges to the prevention and treatment of kidney diseases in the region. By contrast, the relatively low facility costs in many parts of Asia and the abundance of patients and biomedical samples present invaluable research opportunities to address knowledge gaps that are critical in the strive to continue to improve clinical outcomes.



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